Prediction of Chronic Disease Risk in Older Adults using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Background: Chronic disease and functional decline are major contributors to morbidity in older adults, yet the predictive contribution of accumulated self-reported health burden indicators across longitudinal transitions remains underexplored.

Methods: A longitudinal machine learning framework was developed using five waves of TILDA data. Variables were standardised through a multi stage preprocessing pipeline, and four model families (`Logistic Regression`, `XGBoost`, `Random Forest`, `Multi-Layer Perceptron`) were evaluated across three modelling designs spanning single and pooled transitions. Performance, generalisation, and predictor contribution were assessed using Cross Validation (CV), bootstrapped confidence intervals, and predictor add-on analysis.

Results: Pooled multi wave modelling improved predictive performance and stability over the single wave transition baseline, increasing CV AUC from 0.589 ± 0.072 to 0.718 ± 0.006. Reducing the feature set to the 20 most important predictors caused no measurable performance loss, while cognitive, psychosocial, nutritional, and behavioural add-on variables produced no statistically significant improvement.

Conclusions: Chronic disease risk prediction in this cohort using the publicly accessible TILDA dataset was driven primarily by accumulated self-reported and functional health measures rather than isolated domain specific risk factors. A compact set of survey derived indicators captured most predictive signal for risk assessment.

Keywords: ageing, Machine Learning (ML), predictive modelling, chronic disease, functional decline, health burden, The Irish Longitudinal Study on Ageing (TILDA), Extreme Gradient Boosting (XGB)

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AI Declaration

Declaration I acknowledge the use of [AI tool or technology name] and [link] to generate.../ I have not used any AI tools or technologies to prepare this assessment. Prompt: I entered the following prompt/s... Use: I used the output to.../I modified the output to...

Example 1:

I acknowledge the use of [INSERT GEN AI PLATFORM] to generate an outline for an essay. I entered the following prompt: “Provide an outline for a 3000-word essay on the establishment of the University of Limerick.” I used the output at the initial stage of the assessment task to help plan my essay. I modified the outline generated, discarding several suggested body paragraphs and replacing them with my own ideas based on the research and reading I completed.

Example 2

Declaration I acknowledge the use of [INSERT GEN AI PLATFORM] to brainstorm topics for an assessment. I entered the following prompts: “Brainstorm ideas around the topic of the establishment of the University of Limerick.”I used the output as a starting point for initial research before narrowing down the topic for my assessment.

Limerick, 2026

Limerick, 2026

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [7–9]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [10]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [11]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland's *Healthy Ireland* framework, highlight nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [12], Physical Activity (PA) [13], Scales and Questions on Mental Health and Wellbeing (MH) [14], and Social Variables (SOC) [15] improve predictive performance. Estimating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

The Irish Longitudinal Study on Ageing (TILDA)[1] provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced Research and Development Corporation (RAND) Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutrition factors, and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

[1]TILDA's dataset are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to University of Limerick (UL) S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [12, 16].

- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as Frailty (FR) progression [6]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$.

  This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:**Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [17].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).

- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.

- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.

- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and other MH features complement established clinical measures to enhance predictive performance.

- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.

- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis outline

Chapter 2 presents a literature review on ML prediction of chronic disease in ageing populations, focusing on lifestyle, nutrition predictors and methodological considerations. *Chapter* 3 describes the data preparation process, including the TILDA study design, data collection, thematic domain structure, and the manual variable filtering workflow. In *Chapter* 4, the automated cleaning pipeline is presented, including the variable scenario detection system, the deterministic six-step cleaning sequence, and validation of the pipeline against established benchmarks. *Chapter 5* presents the experimental results, including model comparisons and feature importance analysis. *Chapter 6* discusses the implications and future directions of the research.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes.

2.1 Search strategy and study selection

Several methodological papers describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. This review followed key principles for systematic mapping. This included defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined [1]. The integrative literature review [18] helped structure and connect findings across studies, while [19, 20] guided the organisation of ideas. In related methodological work, [2] clarified how Systematic Review (SR)s can identify research gaps, and [21] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing in AI, and health research.

2.1.1 Database search and query development

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as "Irish Longitudinal Study on Ageing", "machine learning", "chronic disease", "nutrition", "diet", and "physical activity", as shown in Table 2.1.

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

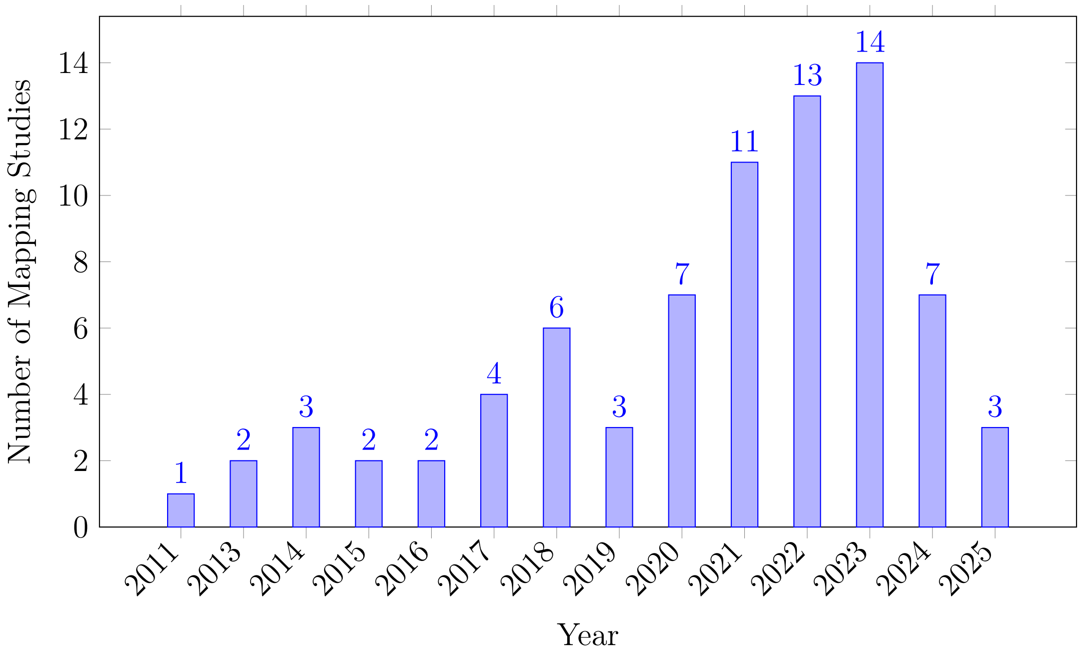

Figure 2.1: Publications per year.

Some search strings returned few or irrelevant entries, particularly for ML applications in TILDA, highlighting a key research gap. Iterations continued until suitable combinations consistently retrieved relevant studies. Duplicates were removed using Zotero's reference management feature. The remaining studies were individually and manually checked for relevance to "ageing", "chronic disease", and ML. Using Zotero's notes functionality, the initially created Table 2.3 captured

Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing) AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	9

key elements for each study, including model type, features used, and main findings, as shown in the following Preferred Reporting Items for Systematic reviews and Meta Analyses (PRISMA) flow diagram (Figure 2.2).

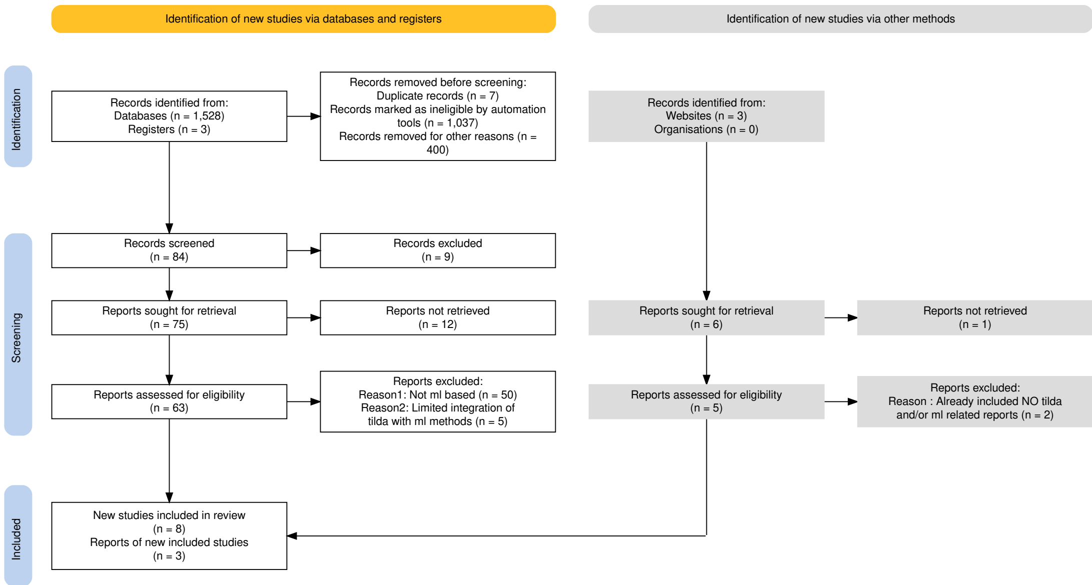

Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [5].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customised version of the Newcastle-Ottawa Scale (NOS) was applied (Table 2.2). The NOS evaluates studies across three domains: *selection* (S1–S4), *comparability* (C1–C2), and *outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored $0/1$, recall the Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: 1.studies critically related to TILDA, 2.studies using TILDA and ML for predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of Systematic Literature Review (SLR)s [2].

Reference	**S1**	**S2**	**S3**	**S4**	**C1**	**C2**	**O1**	**O2**	**O3**	**Total**
(Xu, 2023 [22])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [17])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [6])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [7])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [23])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [24])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [25])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [26])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [27])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [28])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [29])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by *DoctoralWriting* resources [30, 31], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the com-

Table 2.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
Frailty (FR)	Romero-Ortuno (2021) [26]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion (EX), Unexplained weight loss (WL), Weakness (WK), Slowness (SL), Low physical activity (LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + Machine Learning (ML)	Donoghue (2023) [17]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were $\leq$ 0.69 and PRAUC $\leq$ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [32]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with Fear of Falling (FOF) had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [33]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [34]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [16]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [12]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [27]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational (short follow up (2 years)); limited generalisability >65 years, self reported
ML	Xu (2023) [22]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year Diabetes Mellitus (DM) prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [7]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [28]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [23]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

plexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML applications with TILDA, additional studies on older adult populations and chronic disease were included.

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) is a conditional probability $P(Y \mid X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j, guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [26]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [35]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} \mid \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Age also plays an important role: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [33].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X.

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been tested on the TILDA dataset, and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling more robust train-validation splits in ML models, and exploring richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [36] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [37] reported that self reported measures may exaggerate such associations, raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [13]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, mediated by FOF [34]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited, especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [12] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [38] found vitamin D deficiency predicted DM, musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes, which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [27] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [39, 40]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [22, 23]. Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables.

A recurring pattern, however, is that models often over-rely on easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [32]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes. Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.1 Models used in machine learning

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [41] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an AUC of 0.78, whereas ensemble methods in [22] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.2 Performance and validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. Furthermore, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [32].

2.5.3 Explainability in machine learning for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 7, 22]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.6 Gaps in the literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (`50+`). This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA studying ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [12], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [27] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [41] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [22] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where ensembles or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [12] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [7] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [12], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [26] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through listwise deletion, as in [41], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model's outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [4, 22], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, nutrition, and biological ageing. We also examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the data preparation and pipeline implementation described in Chapters 3 and 4, and the modelling strategies presented in Chapter 5.

Chapter 3

Methodology

3.1 Study population and design

TILDA provides longitudinal data across five waves (Waves 1–5) for community dwelling adults in Ireland. For benchmarking against international cohorts, an external harmonised TILDA dataset was also inspected and prepared. The harmonised dataset, which has been pre aligned to international standards e.g., the Gateway to Global Ageing Data (G2G), contains a subset of variables mapped to a common cross study format and thus differs from the raw TILDA files. Although the harmonised dataset was simpler to process, it was still subjected to the variable filtering and numerical standardisation steps developed as part of this research to ensure compatibility with the five wave cleaned datasets prior to modelling.

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering occurred primarily at the *variable level* during the manual screening stage, which excluded features with extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only if all retained variables were missing for that individual after manual variable screening.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications, differing from exclusion heavy cohort definitions often used in traditional epidemiological studies [42].

3.2 TILDA data collection and screening

TILDA data collection occurred over five longitudinal waves (Table 3.1).

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.
2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.
3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centres or at home (Wave 1 and Wave 3).

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Table 3.1: Dimensions of raw TILDA and harmonised datasets version 5.5 (prior to cleaning in Section 4.1).

Dataset	Collection Period	Participants (N)	Raw Variables (N)
Wave 1 [43]	2009–2011	8501	1619
Wave 2 [44]	2012–2013	7206	724
Wave 3 [45]	2014–2015	6397	1029
Wave 4 [46]	2016	5713	990
Wave 5 [47]	2018–2019	4978	983
Harmonised [48]	2016 (release)	8501	572

3.3 Thematic domains

Variables were organised into seven thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the Pseudonymised Microdata File (PMF) metadata (Table 3.2), indicates its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and machine learning.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code (e.g., `ph121`). Variables with multiple responses or loops include an underscore and number to specify the response or iteration (e.g., `ph201_05`).

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area (Table 3.2). These prefixes provide a consistent framework for categorising features and guiding the assignment into thematic domains for downstream analysis.

By categorising the raw variables into these seven thematic domains, a structured foundation was established for the technical dimensionality reduction and cleaning pipeline described in the following Section 3.4 and Chapter 4.

3.4 Manual variable filtering workflow

Prior to any data cleaning or modelling, a structured multi-stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. As shown in Table 3.3 (for example, 1619 → 809 variables in Wave 1), the raw PMF files were reduced to a subset of variables of interest. This procedure also separated automated diagnostic profiling from manual variable exclusion, ensuring transparency in all selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (`.sav`) files, while the harmonised dataset was loaded from a (`Stata`) (`.dta`) file due to technical compatibility issues.

2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

Table 3.2: PMF v5.5 derived variable prefixes [3].

Variable Prefix	Research Area
1: Socio Demographic & Context	
INC	Income variables
SES	Social class
SOC	Social
2: Physical Health & Clinical Baseline	
BP	Blood pressure from sphygmomanometer
MD	Medication data
SCR	Screening tests
3: Functional Status & Frailty	
ADL	Activities of daily living
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
GRIP	Grip strength
IADL	Instrumental activities of daily living
4: Lifestyle & Behavioural	
BEH	Health behaviours
5: Nutrition & Metabolic Biomarkers	
BLOODS	Results of blood sample analysis
6: Psychological & Cognitive	
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
MH	Scales and questions on mental health and wellbeing
Other / Administrative	
G2G	Gateway to Global Aging Data

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds described below. Variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.

4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained `variables of interest`.

5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No cleaning, recoding, or imputation was applied at this stage.

Python scripts were used to profile all variables across each wave. For every variable, the following metrics were computed: data type, number of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format to support structured manual review prior to variable dropping.

Manual filtering was applied according to the following criteria:

- **Missingness:** Drop variables with $> 80\%$ missing values. This threshold balances retaining sufficient feature coverage while ensuring statistical reliability. Variables exceeding this threshold provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.

- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.

- **Dominant category (top value/frequency):** Variables where a single category (sample value) marked as the *topValue* occurs over $\approx > 90\%$

(*topFrequency*) of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.

- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave's PMF file into Comma Separated Values (CSV) files (Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

During manual screening and filtering, variables exceeding exclusion thresholds (e.g., high missingness, single value, dominant category, administrative identifiers) were visually flagged (e.g., marked in red). Non flagged variables were programmatically concatenated with double quotes and comma to seperate each variable using spreadsheet formulas (e.g., `=TEXTJOIN(", ", TRUE, CHAR(34) & J2:Jn & CHAR(34))`) ready for the manual transfer into wave specific JSON files for each wave. These files represent the *variables of interest* retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing and cleaning in Section 4.1.

The retained variables defined in JSON files were subsequently used to subset the raw datasets. As shown in Table 3.3, this process yielded a significant reduced dataset for each wave. For instance, in Wave 1, manual filtering reduced the feature space from 1,619 to 809 variables, a 50.03% reduction. When considering all five waves collectively, the total number of variables was reduced from 5,345 to 3,099 representing a count of 2,246 dropped features with an overall reduction average of 42.0%. This substantial drop highlights the elimination of non-contributory data (providing no predictive value to the model), ensuring only the *variables of interest* proceed to the automated cleaning engine.

Importantly, all variable exclusion decisions were completed at this stage. Subsequent preprocessing steps focus exclusively on numerical standardisation, recoding, and preparation for machine learning rather than further feature removal. With the variables of interest selected and the datasets reduced to a

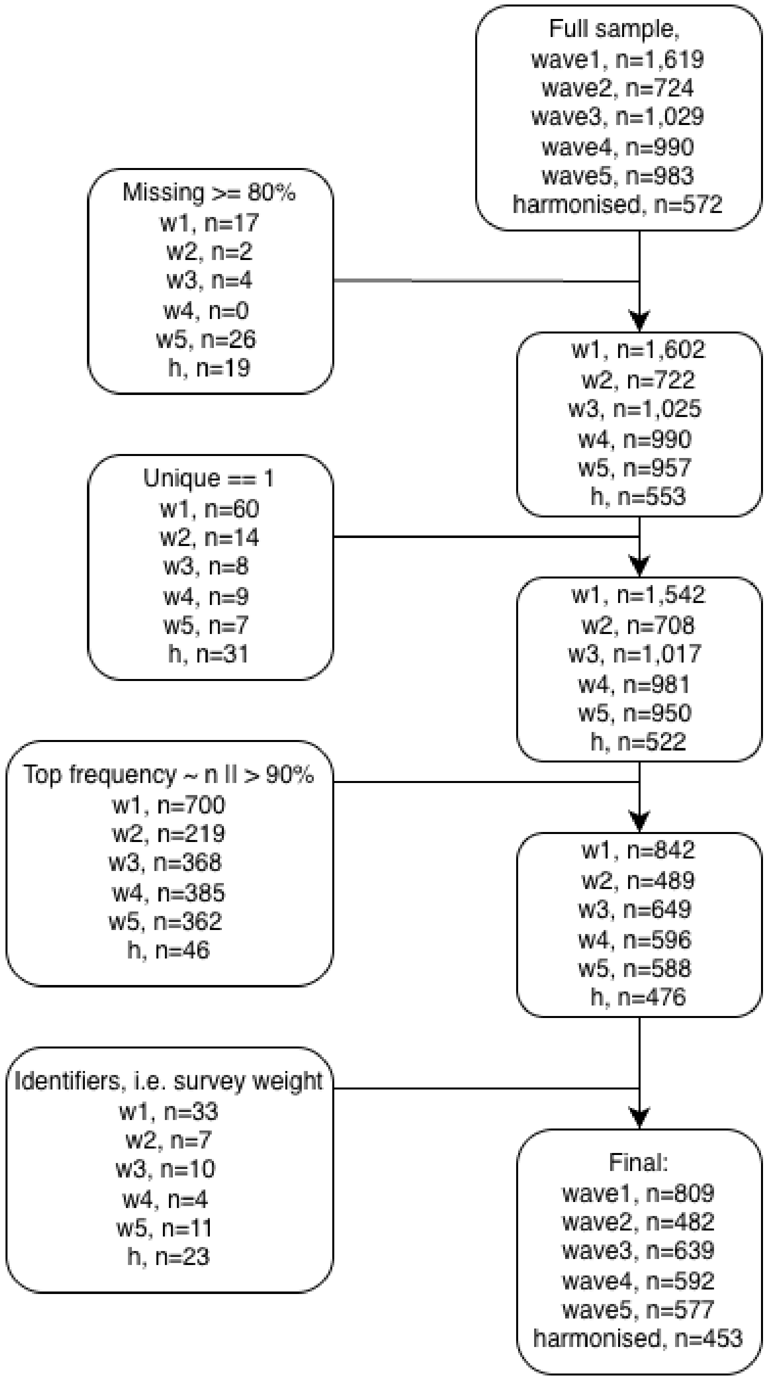

Figure 3.1: Flowchart of the manual five-step workflow for variable screening and filtering. Placeholder (n) values represent the number of variables at each stage, illustrating the reduction from the raw dataset to the filtered dataset prior to automated preprocessing.

manageable dimensionality, the next stage focuses on automated preprocessing, recoding, and preparation for ML modelling.

3.5 Reduced dataset overview

To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`.ipynb`) file implemented with `pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves. The purpose of this profiling stage was not transformation but to expose structural inconsistencies and latent formatting issues that would undermine direct numerical conversion.

Table 3.3: Input shape for cleaning pipeline (post manual filtering workflow 3.1).

Dataset	Row (N participants)	Column (N features)	% Reduction
Wave 1	8501	809	50.0%
Wave 2	7206	482	33.4%
Wave 3	6397	639	37.9%
Wave 4	5713	592	40.2%
Wave 5	4978	577	41.3%
Harmonised	8501	453	20.8%

For example, in wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns. Age distributions confirmed that the study population was restricted to older adults across waves, however, ages were bottom coded for participants younger than 50 years and top coded for those aged 65–85 years and above. Missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

Despite this, the exploratory visualiser was limited because the raw data were not yet cleaned, many outputs were incorrect or nonsensical due to hybrid formats and inconsistent coding. For instance, inspection revealed that many variables across both the wave specific and harmonised datasets, while appearing numeric at first glance, in fact contained hybrid representations combining numeric prefixes with descriptive text (e.g., “1.Rarely or none of the time (less than 1 day)”, “4.All of the time (5–7 days)”, or prefixed labels such as “0.not coupled”, “1.coupled”). A trivial string stripping approach would therefore produce incorrect numeric values (for example, reducing “5–7 days” to `5`) resulting in silently introducing semantic distortion. Similarly, numeric columns sometimes contained sentinel codes (e.g., -98, -99) or embedded symbols (‘$’, ‘*cm*’, ‘%’) that skewed visual summaries.

These insights, revealed through systematic profiling, guided the development of the subsequent cleaning pipeline steps, ensuring consistent numeric logic representations (e.g., midpoints for ranges, standardised sentinel handling).

3.6 Retained variable distribution by domain

Using the PMF prefixes described in Table 3.2 as a guide, variables were assigned to seven thematic groups (Table 3.4), each capturing a conceptually coherent domain: physical, cognitive, social, and behavioural aspects of ageing. This structure supports reproducible preprocessing and helps machine learning models focus on meaningful, longitudinally robust features.

Specifically, the thematic grouping:

1. Provides a clear framework for automated cleaning and filtering rules.
2. Ensures that model input variables are conceptually aligned with study objectives.

The seven thematic groups integrate information from CAPI, SCQ, and derived variables, covering sociodemographic context, physical health, functional status, lifestyle behaviours, nutrition and metabolic biomarkers, psychological and cognitive measures, and administrative identifiers.

The gradient *heatmap* integrated at the bottom of Table 3.4 visually illustrates the relative dominance of each thematic domain across the five waves. Darker blue shading indicates domains with a higher number of variables, providing a quick summary of domain coverage and prominence. This integrated visualisation supports interpretation by highlighting which domains contribute the most features to the dataset.

Table 3.4: Thematic domain grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variable Examples	Total Features
1	**Socio-Demographic & Context**	Baseline demographic and socioeconomic indicators	Age/Sex Education Income Marital status	844
2	**Physical Health & Clinical**	Clinical and self reported health measures	Blood pressure BMI Chronic disease status Self rated health	839
3	**Functional Status & Frailty**	Physical functioning and frailty markers	Activities of Daily Living (ADL)/Instrumental Activities of Daily Living (IADL) Grip strength Gait speed	174
4	**Lifestyle & Behavioural**	Health behaviours and activity patterns	Physical activity Smoking Alcohol use	120
5	**Nutrition & Metabolic**	Metabolic and biomarker indicators	C-Reactive Protein (CRP) Creatinine Lipid markers	22
6	**Psychological & Cognitive**	Cognitive performance and mental health	Mini-Mental State Examination (MMSE) Depressive symptoms Stress scale	291
7	**Other / Administrative**	Survey metadata and structural variables	Sampling weight Completion status Contact names	809

	SocioDemographic_Context	PhysicalHealth_Clinical	FunctionalStatus_Frailty	Lifestyle_Behavioural	Nutrition_Metabolic	Psychological_Cognitive	Other_Administrative	Total_Features
wave1	272	149	48	58	10	97	175	809
wave2	119	148	18	21	0	38	138	482
wave3	164	184	44	14	8	93	132	639
wave4	161	186	30	13	4	36	162	592
wave5	128	172	34	14	0	27	202	577
Total_per_Domain	844	839	174	120	22	291	809	3099

Thematic domain dominance across waves (higher n features per domain).

Chapter 4

Technical Implementation

4.1 Automated cleaning engine

The core challenge in converting heterogeneous variable types across the six datasets (Table 3.3) was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained features using a single numerical representation (`float64`) with a unified missing or non response sentinel (`-1.0`).

The exploratory profiling across all waves showed that categorical variables were dominant overall ($2,916$ columns), while numeric variables were fragmented across multiple types, with `float64` being the most prevalent native numeric representation (462 columns), motivating its use as a common standard to ensure consistency and compatibility with downstream ML models, while accommodating decimal values arising from range midpoints, frequency normalisation, and ordinal scaling that cannot be reliably represented using integer types.

To address this, the solution was implemented as a two stage system. The first stage performs automated *variable scenario detection*, identifying the structural form of each variable based on observed value patterns using the function `detect_column_scenario_mapping` (Appendix C). This stage includes additional diagnostic categories used during development to surface unhandled edge cases.

The second stage applies a deterministic *six step numerical cleaning engine* (Appendix E). While nine scenario categories were initially defined during development, three served exclusively as validation fallbacks and were progressively eliminated as detection coverage improved, with a full scenario to step mapping counts provided in Appendix B. The final pipeline therefore consists of six active transformation steps, applied sequentially and uniformly across all waves.

4.1.1 Stage 1: Variable scenario mapping and validation

Stage 1 focused on automatically identifying how each variable should be handled during cleaning. The automated detection function in Appendix C was developed to classify variables based on their observed value structure, including the presence of sentinel codes, numeric annotations, ordinal labels, and mixed string numeric formats. While multiple intermediate detection categories were used during development, these served solely as internal routing logic. Although all variables passed sequentially through the six step cleaning pipeline, most were fully resolved in the earlier steps (Steps 1–5). The final step (Step 6 in Appendix D) was applied only to residual variables that could not be deterministically resolved by earlier transformations.

The additional detection categories (e.g., "single value categorical", "special symbol mixed", and "unknown/complex") were intentionally included as diagnostic fallbacks to surface unhandled edge cases due to the complexity and size of the data. As rules matured, variables initially defaulted to these fallback categories were progressively resolved into earlier, well defined scenarios, leaving none remaining for the final pipeline execution.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of feature states before and after cleaning, including all representative sample values for each variable. This allowed direct inspection of how the same variable was represented across waves prior to cleaning, particularly in cases where identical semantic categories appeared in different orders or formats. By comparing these grouped samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the `show_grouped_vars_for_scenarios` function (Appendix F) was formatted to create an automatic friendly ".md" file for rapid, surface level verification, which honed the final mapping dictionary (an example can be found in Appendix G) for all features across all waves. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.1.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed sequentially by the *six step cleaning engine*, resolving simple and unambiguous cases first to maximise the proportion of clean numerical features before tackling more complex structures, particularly censored and multi class scenarios (Figure 4.1). With later steps progressively shifting from cleaning into semantic feature engineering, this deterministic sequence applied identical transformation and feature encoding logic to each wave, with later steps (particularly Steps 4–6) performing rule based feature engineering by deriving semantically meaningful numerical representations directly from observed category tokens at runtime. The methodological rationale and implications of this integrated engineering which reframes cleaning as semantic feature construction are detailed in Section 4.1.3. (See Appendix D for implementation).

This design reflects the empirical diversity of survey encodings, where a small subset of variables only reveal their semantic structure after simpler transformations have been applied.

The final cleaning pipeline consisted of six deterministic steps applied uniformly across all TILDA waves, with increasingly complex transformations handled in later stages. Steps 4–6 required particular attention due to hybrid, ordinal, or multi class formats. A summary of each step and examples is provided below.

1. **Empty / Blank Values:** All missing entries, empty strings, and whitespace were converted to the standard non-response sentinel (`-1.0`).

2. **Sentinel Values:** Wave specific non-response codes (e.g., `-98`, `-99`) and textual sentinels (e.g., `‘‘dk’’`, `‘‘not applicable’’`) were mapped to the unified (`-1.0`) placeholder.

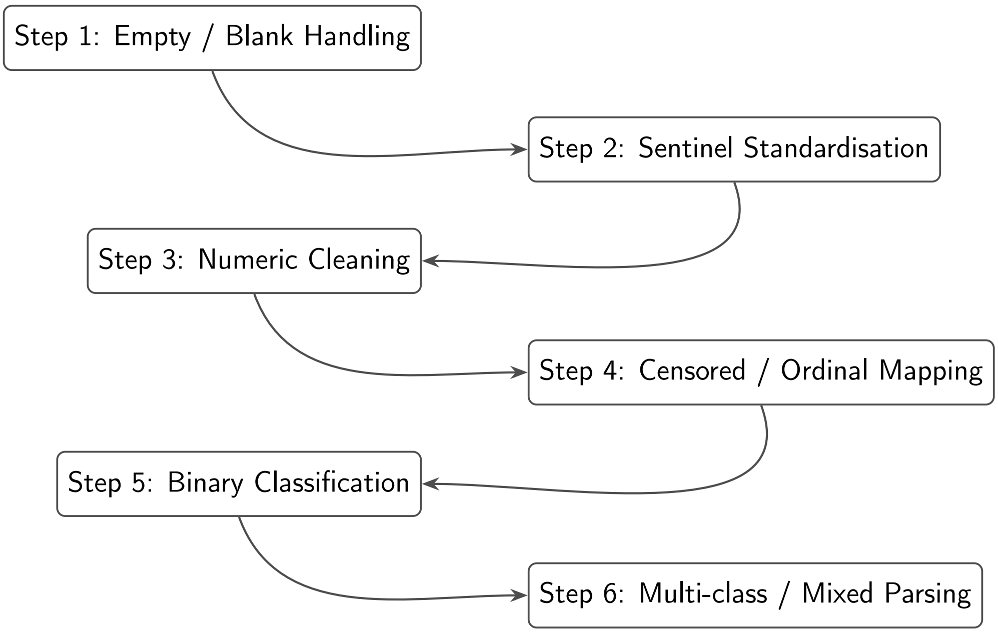

Figure 4.1: Flowchart of the final six-step cleaning pipeline applied to all TILDA waves. Detailed transformation tables are provided in Appendix B.7, and the corresponding implementation "`.ipynb`" script is documented in Appendix E at each scenario step.

3. **Numeric Transformation:** Numeric like columns containing embedded units, symbols, or hybrid strings (e.g., `50+`, `<10`, `1-2.5`) were parsed using regex and converted to `float64`. For ranges, midpoints &/or boundary values were used. For censored or annotated values, appropriate upper/lower bounds were applied.

4. **Censored / Ordinal Variables:** This step addressed variables representing ordinal scales, bracketed numeric ranges, or frequency expressions (e.g., “2–3 days a week”, “daily”). Transformations included:

   - Converting textual ordinal labels and Likert scale responses to numeric equivalents.
   - Mapping calendar based frequency descriptors to numeric weekly equivalents.
   - Manual verification for low count or unusual variables, supported by the Markdown audit outputs (Appendix G), to ensure semantic meaning was preserved.

5. **Binary Categorical Variables:** Two level features (e.g., Yes/No, Male/Female) were dynamically encoded using mappings derived from the observed value set of each variable:

   - Semantic rules were applied e.g., (“no”, “not”, “female” $\rightarrow$ 0.0) and (“yes”, “male” $\rightarrow$ 1.0).
   - Count based detection verified that variables had exactly two non-missing unique values.
   - Features with more than two sample values were also analysed for sentinel presence:
     - If a variable had three observed values and one was a sentinel, it was treated as binary.
     - If a variable had four observed values and two were sentinels, it was treated as binary.
   - Sentinels were preserved as `-1.0`.

- This approach ensured order invariant encoding across waves, even when original labels or numeric prefixes differed.
- *Example (sentinel case):* Feature `fl002_5` with pre transformed values `['NOT Getting in or out of bed', -1.0, 'Getting in or out of bed']` was encoded as `[0.0, -1.0, 1.0]`, demonstrating how a sentinel preserves binary semantics.

6. **Multi-class / Mixed Variables (rule based feature engineering):** Variables with remaining multi class structure, mixed types, or unresolved string numeric hybrids after earlier transformations were processed in the final step.

   Technical implementation details for this step are provided in Appendix D:

   - Lexical parsing extracted numbers, ranges, units, percentages, and textual frequency descriptors into floats, yielding interpretable numerical feature representations.
   - Sentinel handling ensured all known non-response codes mapped to (`-1.0`).
   - Original values were grouped and assigned numeric identifiers in first seen order, non numeric categorical labels were assigned integer codes above the highest observed numeric value to prevent collisions.
   - Manual inspection using the Markdown audit file verified that semantic equivalence was maintained across waves.
   - Step 6 was designed to be *order invariant*, dynamically interpreting category values within each wave rather than relying on predefined lookup tables shared across waves. This preserved consistent numeric representation regardless of variable ordering or wave specific encodings.
     - *Example:* Step 5 handled simple binary variables such as `CHRany_pain` (any pain reported) with pre transformed values `['pain', 'No pain']` and numeric mapping `[1.0, 0.0]`. In contrast, Step 6 dynamically processed multi class variables like `we114` (pre tax

wage/salary) many of which contained over 20 distinct pre transformed categories (e.g., `-1.0, '200-299', 'Less than 100', ..., '6,000-6,999'`) into consistent numeric representations (e.g., `-1.0, 249.5, 99, ..., 6499.5`). This illustrates why Step 6 required wave specific, dynamic parsing to maintain semantic equivalence across diverse observed values.

4.1.3 Feature engineering within the cleaning pipeline

Although the six step pipeline is framed primarily as a numerical cleaning procedure, Steps 4–6 also perform explicit rule based feature engineering by interpreting heterogeneous categorical survey responses and converting them into semantically meaningful numerical representations. Rather than acting as a separate post processing stage, feature engineering is embedded directly within the cleaning logic, formalising ordinal structure, frequency, and magnitude encoded in text into directly modelable `float64` values suitable for downstream analysis.

Crucially, this process engineers the *representation* of each feature rather than expanding the feature space. For example, free text frequency responses such as "`once or twice a week`", "`a couple of times per month`", or "`almost every day`" are not treated as nominal categories. Instead, lexical cues (e.g., *once*, *twice*, *couple*) are resolved to numeric quantities and normalised onto a common temporal scale, yielding a continuous rate representation. Implicit semantic information such as ordering, intensity, and frequency is therefore preserved through deterministic parsing and normalisation without introducing additional variables. This design maintains the original dimensionality of the dataset while ensuring numerical homogeneity and semantic integrity, producing a compact and model ready representation of complex real world survey data.

4.1.4 Cross wave semantic verification

Given the absence of guaranteed category ordering across waves, preserving semantic equivalence during numerical encoding and feature representation was essential to avoid introducing distortions into downstream ML modelling. Pre and post transformation values for each group were exported to a readable Markdown (`.md`) file, as shown in Appendix G. This representation confirmed that features of known importance for downstream predictive models were preserved and correctly encoded. For example, features such as `ph001` (self rated general health), `ph009` (comparative self rated health), and `bh006` (smoking frequency) correspond to items identified by [6] as among the most influential predictors in mortality models by the Frailty Index (FI). Retaining these features from the manual filtering thresholds applied in (Section 3.4) and encoding them consistently across waves ensured that the cleaned dataset maintains coverage and semantic integrity, preventing distortions that could potentially compromise model performance.

A key challenge in cleaning the five TILDA waves was the lack of guaranteed ordering for categorical values of the same variable across different waves. Table 4.1 illustrates this for waves 4 and 5.

For example, gender (`sex`) variables appeared under different value encodings, with varied category orderings and numeric prefixes across waves. Values such as `['Male', 'Female']`, `['Female', 'Male']`, and prefixed forms like `[''1.male', '2.female']` were all mapped to a consistent numeric representation (`1.0` for male, `0.0` for female). This ensured that semantic equivalence, rather than positional ordering or wave specific encoding, determined numerical assignment.

The variable `ph001` found in Appendix G corresponds to the survey question: "Now I would like to ask you some questions about your health. Would you say your health is…"?. Responses are ordinal (Poor to Excellent) with non-response categories.

The grouped Markdown representation reveals three critical points:

1. Semantically identical categories are correctly grouped even under different variable names (`ph001` vs `ph009`), enabling direct comparison of representations that would otherwise appear fragmented across waves.

2. Numerical encoding is invariant to category ordering, preserving ordinal meaning across waves. Without this approach, positional encoding would

assign inconsistent values, potentially distorting performance for downstream models.

3. For variables with a large and heterogeneous set of observed values (e.g., income bands, frequency ranges, or mixed numeric text responses), the grouped view provided essential visibility into how complex category spaces were being collapsed into stable numeric representations, supporting both validation and error detection at scale.

Table 4.1: Inconsistent category ordering for the same feature *ph*009 (`self rated health compared to other people your age`) across waves and its impact on positional (order based) label encoding.

Pos	**Wave 4**		**Wave 5**	
	Category	**Code**	**Category**	**Code**
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

4.2 Validation on the cleaning pipeline

The purpose of the cleaning validation stage was to assess whether the automated numerical parsing and encoding pipeline preserved meaningful biological and statistical structure within the cleaned datasets. Rather than asserting correctness based solely on successful execution, validation was performed by evaluating whether known relationships reported in the literature could be recovered using the cleaned data under realistic data governance constraints.

Due to restricted access to certain outcome variables within TILDA, full end-to-end replication of all published models was not always feasible. In particular, access through the `Hotdesk Facility` for the official mortality outcomes is subject to additional governance requirements, including institutional GDPR training certification and approval by the TILDA data access committee [49].

As a result, validation focused on two complementary strategies:

(i) Internal consistency checks against established biological and clinical relationships, and

(ii) Constrained replication of published modelling frameworks using available predictors and the derived `8y_mortality_proxy` outcome.

Together, these validation steps demonstrate that the cleaning pipeline retains expected relational patterns and produces datasets appropriate for downstream modelling tasks.

4.2.1 Internal feature relationships

The first validation stage assessed whether cleaned predictors preserved biologically and clinically plausible relationships reported in prior TILDA studies. This analysis focused on predictors identified by [4] as accounting for approximately 95% of the explained variance in Gait Speed Reserve (GSR), defined as the difference between maximum and usual gait speed.

As GSR is not directly available in the public wave 3 dataset, validation was performed by examining the directional relationships between the 7 primary predictors themselves. This approach evaluates whether the numerical encoding process, including sentinel handling, missing value treatment, and ordinal mapping, retained the expected internal structure of the data rather than reproducing outcome specific performance.

Figure 4.2 presents the correlation matrix for the selected predictors following cleaning. Grip strength exhibited a negative association with both age and chair stand time, reflecting the inverse relationship between muscular strength and physical performance decline. Cognitive performance (Montreal Cognitive Assessment (MOCA)) demonstrated a moderate negative correlation with age and a positive association with educational level, consistent with known patterns of cognitive reserve in older adults. Education correlated positively with cognitive performance, and BMI showed the expected weak negative association with age, reflecting reduced physiological reserve in older cohorts.

Table 4.2 summarises the expected and observed directional relationships between key predictor pairs. Notably, grip strength showed moderate negative correlations with both age ($r = -0.265$) and chair stand time ($r = -0.170$), reflecting the expected inverse relationship between muscular strength and physical performance decline. Cognitive performance (MOCA) displayed a moderate negative association with age ($r = -0.381$) and a positive association with education ($r = 0.367$), consistent with known patterns of cognitive reserve in older adults. BMI demonstrated the anticipated weak negative correlation with age ($r = -0.073$). These results confirm that the cleaning pipeline retained the expected numerical structure and relational patterns of the predictors, rather than introducing artificial associations.

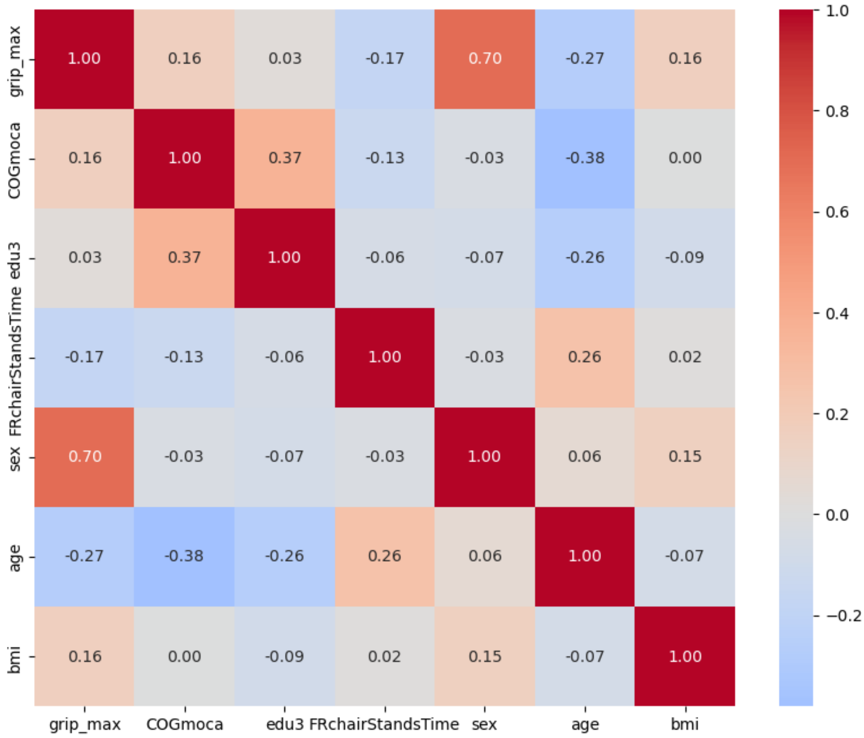

Figure 4.2: Correlation heatmap for top 7 predictors (Wave 3).

Table 4.2: Directional validation of the 7 features explaining 95% of Gait Speed Reserve (GSR) variance as identified by [4].

Predictor 1	**Predictor 2**	**Expected Direction / Justification**	**Observed r**
Grip Strength (GRIP)	Age	**Negative (−)**: GRIP is a positive predictor of GSR, while age is a negative predictor.	-0.265
GRIP	Chair Stand Time	**Negative (−)**: Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests.	-0.170
Montreal Cognitive Assessment (MOCA) Score	Age	**Negative (−)**: Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve.	-0.381
Education	MOCA Score	**Positive (+)**: Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR.	0.367
Body Mass Index (BMI)	Age	**Weak / Mixed**: BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak.	-0.073

4.2.2 Model based validation

To evaluate whether the cleaned dataset preserves predictive utility, a constrained replication of [6] was conducted. The resulting predictive performance for both model variants and sexes is shown in the following Figure 4.3. The study predicted 8-year mortality using 32 FI items. In the cleaned wave 1 dataset following the deterministic six step cleaning pipeline described in Section 4.1.2, only 16 of these 32 FI items remained following the manual variable filtering described in Section 3.4, which prioritised global numeric consistency across waves.

An '`8y_mortality_proxy`' outcome was constructed to approximate the sex specific 8-year mortality rates reported in [6] (men: 14.9%, women: 11.1%), derived from linkage to Ireland's Central Register Office records [50]. Due to access restrictions, the official mortality data were unavailable. Therefore, mortality status was first assigned by randomly selecting deceased participants from the wave 1 cohort aged 50 years or older to match the published sex specific mortality proportions observed at wave 5 follow up. An equal number of surviving participants were then randomly sampled within each sex to construct balanced datasets for model training. The resulting random proxy datasets included 1,114 men (557 deceased) and 982 women (491 deceased), reflecting the baseline wave 1 population aged ≥ 50 years ($n = 8,172$: 3,743 men and 4,429 women).

Two variants of this proxy were implemented:

(i) the purely random assignment described above, used as a structural baseline, and

(ii) biologically informed probabilistic assignment that leverages latent frailty captured via Principal Component Analysis (PCA). This dual construction allowed separation of statistical artefact from biologically meaningful signal.

The random proxy provides an initial diagnostic benchmark, mortality labels were assigned using purely random sampling within each sex to match the published 8-year mortality proportions. While this preserves overall class balance, it does not encode any biological or frailty related information and therefore serves only as a structural validation baseline.

For the biologically informed proxy, the 16 available FI items were standardised to zero mean and unit variance using z-score transformation. PCA was then performed separately within each sex on complete case observations to extract the principal component representing the dominant latent frailty structure. The First Principal Component (PC1) was retained as a latent frailty score, and its direction was aligned such that higher PC1 scores correspond to higher frailty by enforcing a positive correlation with the calculated FI. This latent frailty score was subsequently combined with age to construct a probabilistic mortality assignment reflecting both biological vulnerability and time dependent risk.

Two model variants were tested for each sex:

1. **FI only model:** using the 16 available FI items, without age.

2. **FI + Age model:** including age as an additional predictor.

Model training and evaluation followed the procedure reported in [6] as closely as permitted by the available data. Features were min-max normalised prior to modelling. For each sex, data was split into training (80%) and testing (20%) sets. To mitigate overfitting and obtain robust estimates of model performance, 5-fold stratified cross-validation was applied on the training data. AUC was selected as the primary evaluation metric due to its widespread use in prior TILDA and ML based studies (Figure 2.2) and its robustness to class imbalance.

For six FI items (`mh0014`, `ph001`, `ph102`, `ph108`, `ph114`, `bh201`), the cleaned dataset transformations differed from the scoring rules published in [6]. To align precisely with the original 16/32-item FI methodology, these items were recoded following the published ordinal and dichotomous mappings. This step was necessary to ensure that the six remapped FI items reflects the same interpretation of health deficits used in the literature and is consistent with standard FI construction principles [51]. Models were then re-evaluated using this partial remapping FI to assess sensitivity to feature encoding.

The resulting AUC values are reported separately for (i) cleaned (Figure 4.3) vs 6/16 recoded FI items (Figure 4.4), (ii) random vs PCA informed mortality assignment, (iii) FI only vs FI + Age models, and (iv) men and women.

These results highlight several key points:

1. **Random proxy baseline (cleaned vs recoded FI):** Under purely random mortality assignment, discrimination remained at chance level for both versions of the FI. For the cleaned 16 FI items, the FI only model achieved AUC = 0.507 (men) and 0.512 (women), increasing marginally to 0.516 (men) and 0.531 (women) when age was included. When six of the sixteen available deficits were recoded to align with the published FI scoring rules [6], the AUC values increased to 0.567 (men) and 0.537 (women) for the FI only model, and to 0.572 (men) and 0.605 (women) for the FI + Age model. Despite this increase relative to the fully cleaned encoding, discrimination remained close to chance under random label assignment.

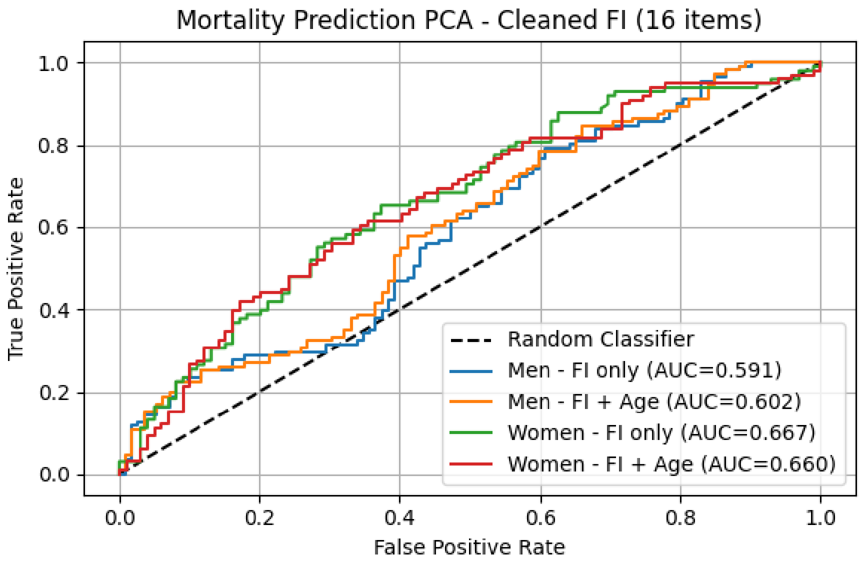

Figure 4.3: ROC curves for mortality prediction using the cleaned 16 FI items under the PCA informed mortality proxy. Separate models are shown for men and women, with and without age.

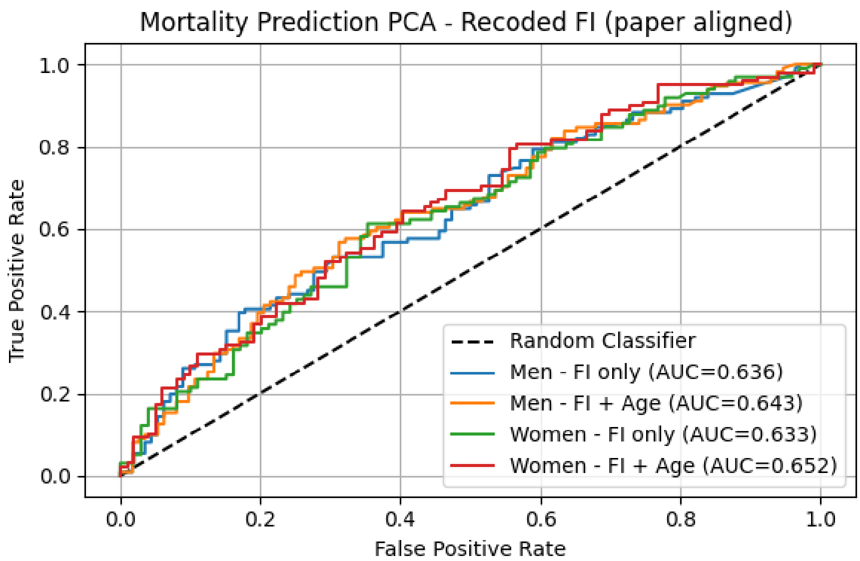

Figure 4.4: ROC curves for mortality prediction using the 6/16 recoded FI items aligned with [6]. Performance is shown separately for men and women, with and without age.

These results confirm that, without embedding biological frailty structure into the outcome, predictive performance reflects statistical noise rather than meaningful signal, regardless of encoding refinement.

2. **PCA informed mortality proxy (cleaned vs recoded FI):** When mortality assignment incorporated the PCA derived frailty representation PC1, discrimination improved relative to the random baseline for both FI constructions. For the 16 cleaned FI items, the FI only model achieved AUC = 0.591 (men) and 0.667 (women), increasing to 0.602 (men) and 0.660 (women) when age was included. When six of the sixteen deficits were recoded in accordance with the published scoring framework, the FI-only model achieved AUC = 0.636 (men) and 0.633 (women), while the FI + age model achieved 0.643 (men) and 0.652 (women).

   This indicates that closer alignment with published ordinal deficit scoring improves discriminative stable performance relative to the simplified numeric encoding applied during initial cleaning.

3. **Comparison with published performance:** Absolute AUC values remain below those reported in [6] (approximately 0.70 for FI only and up to 0.80–0.90 when age is included), which is expected given the use of a constructed mortality proxy and only 16 of the original 32 FI items. However, the methodological sequence reported in the original study is preserved:

   (i) Random assignment yields chance level performance,

   (ii) Extracting latent frailty structure via PCA improves discrimination,

   (iii) Inclusion of age generally improved mortality prediction.

   Importantly, the cleaned dataset maintains realistic sex distributions (45.8% men, 54.2% women) and mortality proportions consistent with published estimates, confirming that preprocessing preserved demographic and structural integrity required for downstream modelling.

Chapter 5

Empirical Evaluation

5.1 Modelling and evaluation framework

This section defines the modelling and evaluation setup used throughout this chapter, including model families and designs, analysis samples, outcome definitions, and evaluation procedures, building directly on the prepared datasets from Chapters 3 and 4.

The evaluation is based on participants aged between 50–84 years with a mean age of 66. To ensure that the prediction task reflects incident (new onset) disease, individuals with the target condition present at the start of a transition were excluded, leaving only at risk observations for training and testing. This rule was applied consistently across all designs and for both Track A (Any disease outcome) and Track B (disease specific outcomes).

Final analysis samples differed by design and tracks due to data structure. The baseline design (D1) uses a single transition (Wave 1→Wave 5), with a follow-up period of approximately ≈8 years. The pooled multi-wave designs (D2 and D3) aggregate all eligible consecutive transitions (W1→W2, W2→W3, W3→W4, W4→W5), each spanning approximately 2 years, thereby capturing shorter term dynamics.

Sample sizes and incident rates are summarised in (Table 5.1). Track B, sample sizes vary by condition. (Appendix H) provides the number of observations and incident rates for each disease specific model.

5.1.1 Outcome definitions (Track A and Track B)

Two complementary prediction tasks were defined.

Track A: Predicting any chronic disease onset. Track A uses a binary outcome indicating the first occurrence of any chronic disease during follow-up.

- In D1, disease free status at wave one was established using a composite of self-reported conditions (e.g., eye conditions, cardiovascular conditions, diabetes related conditions, general chronic conditions, limiting long term illness, and self-reported chronic/health problems). Incident cases were then defined using any positive `ICD10_xx` codes from the WHO (*International Statistical Classification of Diseases and Related Health Problems 10th Revision*) at wave five (e.g., mental and behavioural disorders & musculoskeletal diseases).
  - No single `ICD10_ANY` variable exists in wave 1 to establish a clinical baseline. These clinical codes were recorded unevenly across the middle waves, with only three indicators available in wave 2, two in wave 3, and one in wave 4, before a comprehensive set of 16 variables appeared at wave 5 partially a consequence of the manual low heterogeneity filtering flow described in (Chapter 3.4).
  - Physical & Cognitive Health (PH) variables are self-reported responses that reflect prior doctor diagnoses ("has a doctor ever told you"), whereas the "self-reported long term chronic/health problem" variable captures a general self-assessment of long term illness without requiring clinical confirmation.
  - All feature scoring schemes and labels are provided in (Appendix I).
- In D2 and D3, (self-reported long term chronic/health problem) variable was used to both define the outcome and identify the at risk cohort (y=0) at the start of each transition, as it is available in every wave and enables consistent application across all transitions.

$$\text{Coverage}_y = \frac{\sum_{i=1}^{N} \mathbb{1}\big(y_i \neq missing(-1)\big)}{N} \times 100 = 99.91\%$$

- An analysis of wave 5 data compared long term chronic/health problem against clinical records. While raw agreement reached 59.5%, the (*Cohen's Kappa for agreement beyond chance*) (Kappa: 0.167) indicates poor consistency. This divergence reveals that 45% of individuals who perceived themselves as healthy possessed an objective clinical diagnosis, while 38% of those reporting illness had no corresponding hospital record. These results confirm that "self-reported long term chronic/health problem" serves as a distinct proxy for self-perceived chronic illness status. It captures a subjective signal of health burden that exists independently of formal clinical markers.

Table 5.1: Cohort sample sizes by design for Track A

Design	**Observations (n)**	**Incident cases (n)**	**Incidence rate (%)**
D1: Baseline (W1→W5)	916	248	27.07
D2: Multi-wave pooled	14,296	3,212	22.5
D3: Top-20 (reduced)	14,296	3,212	22.5

Track B: Disease specific prediction. Track B targets individual chronic disease outcomes, e.g., circulatory disease, diabetes, and functional decline. For each prediction task, the outcome was defined as the first recorded diagnosis of the specific condition, and participants were excluded if the condition was present at the baseline of the corresponding transition. Scoring details for each disease specific outcome are also provided in the same (Appendix I).

5.1.2 Modelling designs

Four supervised learning algorithms were evaluated across all main Track A designs (D1, D2, D3) to compare linear and non-linear model families under the same evaluation framework.

1. **Logistic Regression (LR)** — A linear model providing interpretability through coefficients and odds ratios. In D1 and D2, a fixed configuration was used: `solver=‘‘liblinear’’`, `C=0.001`, `class_weight=‘‘balanced’’`,

max_iter=2000. In D3, hyperparameters were tuned via RandomizedSearchCV over C (log-uniform $[10^{-4}, 10]$), solver=[‘‘saga’’], l1_ratio (uniform $[0, 1]$), and max_iter $\in \{1000, 2000, 5000\}$.

2. **Extreme Gradient Boosting (XGB)** — Aka XGBoost, is a gradient boosted tree ensemble capturing non-linearities and feature interactions. In D1 and D2, fixed parameters were used: n_estimators=1500, max_depth=3, learning_rate=0.03, min_child_weight=5, gamma=0.1, subsample=0.8, colsample_bytree=0.7, with class imbalance handled via scale_pos_weight (negative to positive ratio). Early stopping (early_stopping_rounds=50) on an internal 20% validation split determined the optimal number of boosting rounds. The model was then refit on the full training set at that iteration. In D3, all tree and regularisation parameters were additionally tuned via RandomizedSearchCV.

3. **RF** — A bagged ensemble of decision trees providing robustness through random feature subsampling (max_features=‘‘sqrt’’). Fixed parameters in D1/D2: n_estimators=500, max_depth=5, min_samples_leaf=10, with class imbalance handled via class_weight={0:1, 1:neg/pos ratio}. In D3, n_estimators (100–800), max_depth (3–10), min_samples_leaf (5–30), and max_features $\in$ {‘‘sqrt’’, ‘‘log2’’} were tuned via random search.

4. **Multi-Layer Perceptron (MLP)** — A feedforward neural network with two hidden layers capturing non-linear hierarchical representations. Fixed parameters in D1/D2: hidden_layer_sizes=(32,16), activation=‘‘relu’’, solver=‘‘adam’’, alpha=0.01, learning_rate_init=0.001, max_iter=500, with early stopping (n_iter_no_change=20, validation_fraction=0.1). Class imbalance was handled via compute_sample_weight(class_weight= ‘‘balanced’’) passed at fit time, as MLP does not natively support class_weight. In D3, hidden_layer_sizes $\in$ {(32,), (64,), (32,16), (64,32), (128,64)}, activation $\in$ {‘‘relu’’, ‘‘tanh’’}, alpha (log-uniform $[10^{-4}, 1]$), and learning_rate_init (log-uniform $[10^{-4}, 10^{-1}]$) were tuned via random search.

All four models shared the same preprocessing backbone: median imputation (SimpleImputer), variance filtering (VarianceThreshold, threshold $= 0.01$), and a missingness filter that dropped any feature exceeding 50% missing values in the training set. Standard scaling (`StandardScaler`, zero mean and unit variance) was applied only for LR and MLP, which are sensitive to feature magnitude. Tree-based models (XGB and RF) were trained on the raw imputed features without scaling. All preprocessing steps were fit exclusively on the training portion (prior to internal train/validation splitting) before being applied to validation and test sets, preventing any information leakage.

Track B models used XGB exclusively, as it achieved the strongest and most consistent performance across D2 and D3 in Track A (see Section 5.2). The same model configuration and hyperparameter search space used in D3 was applied in Track B.

The three primary modelling designs were implemented to address and assess the effect of baseline prediction, longitudinal pooling, and feature reduction with optimisation, respectively.

- **D1 (Baseline):** A single W1→W5 transition. Only participants disease free at wave 1 (using self-reported and derived disease variables, plus explicit `ICD10` source variables) were included, with incident disease defined by any positive `ICD10_ANY` code at wave 5 (n = 916, incidence 27.1%). Feature selection used `SelectKBest` ($k = 20$) and CV used `StratifiedKFold` (one observation per participant).

- **D2 (Multi-wave pooled):** All four eligible consecutive transitions (W1→W2, W2→W3, W3→W4, W4→W5) were stacked into a single pooled dataset. Disease free status and incident outcome were defined using (self-reported long term chronic/health problems) at each transition. A numeric wave transition indicator (`wave_transition_num`) was added as a feature to capture temporal context. The pooled cohort comprised n = 14,296 person wave observations (incidence 22.5%). Feature selection used `SelectKBest` ($k = 100$), and CV used `StratifiedGroupKFold` to ensure no participant appeared in more than one fold.

- **D3 (Top–20 + Hyperparameter Optimization (HPO)):** Built directly on the D2 cohort and wave structure. The feature set was restricted to the 20 highest ranked predictors by XGB gain importance from D2, plus `wave_transition_num`. No additional feature selection step was applied. HPO was performed for all four model families using `RandomizedSearchCV` (200 iterations) within a `StratifiedGroupKFold` framework.

All three designs shared the same participant-level train/test split (80%/20%, stratified by outcome), the same missingness filter and variance threshold applied on training data only, and the same five fold CV evaluation protocol. Track B models followed either the D2 or D3 structure depending on the outcome being predicted. Full details are provided in Subsection 5.2.4.

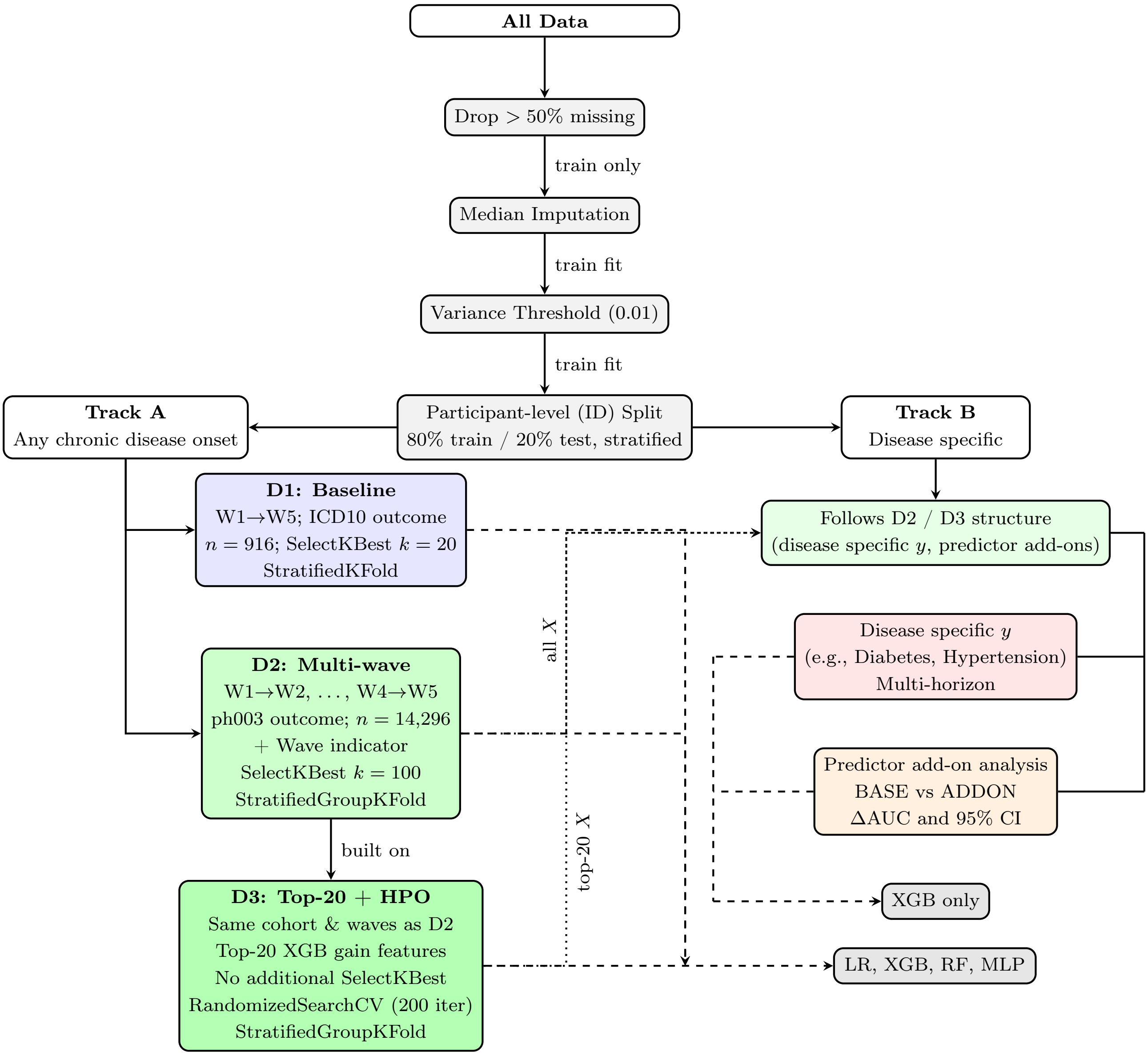

Figure 5.1: Pipeline overview: shared preprocessing applied to all data before branching into Track A (any_chronic_disease, D1/D2/D3) and Track B (disease_specific, following D2/D3 structure). D3 is built directly on D2's cohort and wave structure, adding feature reduction and HPO.

Figure 5.1 notes:

(i) All grey pipeline steps (missingness filter, imputation, variance threshold, split) are shared across Track A and Track B. All fitting occurs on training data only.

(ii) D1 uses a single W1→W5 transition with `StratifiedKFold`, D2 and D3 use all four pooled transitions with `StratifiedGroupKFold` to prevent participant-level leakage across folds.

(iii) D3 is built on D2: it uses the same cohort, waves, and wave indicator, but replaces `SelectKBest` with the top-20 features ranked by XGB gain from D2, and adds HPO for all model families.

(iv) Feature counts after the missingness filter: D1 retains ≈570 features, further reduced to $k = 20$ by `SelectKBest` before model training. D2 retains ≈261 features, reduced to $k = 100$. D3 uses exactly 21 features (top-20 + wave indicator) with no further selection.

(v) Track B uses XGB only, following the D2 or D3 design with a disease specific outcome and explicit exclusion of direct outcome proxies from the predictor set.

5.1.3 Participant-level data splitting and leakage prevention

To prevent identity based leakage where models memorise participant specific health signatures rather than learning general risk patterns, all splitting was performed at the **participant level**. Individuals were assigned exclusively to either the training, validation, or test set, ensuring no participant contributed data to more than one split. The overall dataset was first split into 80% for model development and 20% as a held-out test set. The 80% development partition was then further divided 80%/20% internally, yielding a final three way allocation of 64% training, 16% validation, and 20% test. Splitting was performed using participant identifiers to ensure group wise separation, with stratification by outcome to preserve class balance across splits. Rumala [52] demonstrates in longitudinal brain Magnetic Resonance Imaging (MRI) analysis, this subject wise approach prevents overconfident performance and identity confounding, where models learn individual characteristics alongside diagnostic features, ensuring generalisation to truly unseen individuals.

For D1, where each participant contributes exactly one observation, standard `StratifiedKFold` ($k = 5$) was used. For D2 and D3, where participants may appear across multiple wave transitions, `StratifiedGroupKFold` ($k = 5$) was used to ensure that all observations from a given participant were assigned to the same fold during CV, preventing within participant data from appearing in both training and validation folds simultaneously.

Three additional leakage controls were applied across all designs:

- **Pipeline isolation:** Imputation medians, variance masks, and feature selection scores were computed on the training partition only. The resulting transformations were then applied unchanged to validation and test data.
- **Temporal ordering:** Predictors were drawn exclusively from the baseline wave of each transition (Wave n) to predict outcomes at the subsequent wave (Wave $n + 1$), eliminating any look ahead bias.
- **Exclusion of direct proxies:** Variables that directly encode the outcome being predicted (e.g., self-reported disease source indicators that feed into the `ICD10` codes used as outcome, such as ("has a doctor ever told you that you have") a Ministroke or TIA and Parkinson's disease for nervous system diseases), were removed from the predictor set prior to modelling.

5.1.4 Evaluation protocol and decision metrics

Model performance was assessed using a combination of discrimination, calibration, and classification metrics. Models were trained by minimising their respective objective (loss) functions, while performance was evaluated using out-of-sample metrics. The primary metric AUC, chosen for its robustness to class imbalance and its interpretability in risk prediction contexts. Model performance was estimated using 5-fold CV on the training set. For D2 and D3, this was implemented using `StratifiedGroupKFold` to ensure participant-level separation across folds, while D1 used `StratifiedKFold`. Models that required HPO (D3), a random search (`RandomizedSearchCV`) was conducted within the same CV framework.

Calibration was evaluated using the Brier score, which measures the accuracy of predicted probabilities, while classification performance was summarised using the F1 score, capturing the balance between precision and recall in the presence of class imbalance (e.g., 22–27% incidence). Additional metrics, including sensitivity, specificity, and confusion matrices, were used to support interpretation.

For Track B predictor add-on analysis, 95% Confidence Interval (CI)s for performance deltas were estimated via bootstrap resampling to ensure statistical significance.

5.2 Main results

This section discusses the main results for Track A 5.1.1 and Track B 5.1.1. Performance is first evaluated across the three modelling designs (D1, D2, and D3) 5.1.2, followed by cross design and model level comparisons.

5.2.1 Track A: Any disease onset

Results are presented across the three modelling designs, with tables summarising discrimination (AUC), calibration (Brier score), and classification performance (F1), alongside training, validation, and test metrics to assess generalisation.

5.2.1.1 Design 1 performance:

Table 5.2 summarises the performance of the four models on the single transition (Wave 1 $\rightarrow$ Wave 5). Discrimination is weak with test AUC values ranging from 0.53–0.58, and the larger train-test gaps for tree based models indicate stronger overfitting in the baseline design.

Table 5.2: Predictive performance across model families for D1 (baseline design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.589	0.677	0.582	0.571	0.245	0.411
XGBoost	0.569	0.825	0.563	0.544	0.240	0.398
Random Forest	0.579	0.871	0.565	0.534	0.231	0.440
MLP	0.536	0.664	0.552	0.582	0.241	0.419

5.2.1.2 Design 2 performance:

Table 5.3 shows the results when using all consecutive transitions (pooled multi-wave design). All models improve substantially compared to D1 5.2.1.1, with test AUC values ranging from 0.69 to 0.72.

Table 5.3: Predictive performance across model families for D2 (multi-wave design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.715	0.728	0.715	0.709	0.217	0.460
XGBoost	0.716	0.736	0.719	0.715	0.214	0.468
Random Forest	0.711	0.740	0.717	0.711	0.212	0.452
MLP	0.698	0.749	0.706	0.699	0.208	0.455

5.2.1.3 Design 3 performance:

Table 5.4 presents performance using only the 20 most informative predictors (derived from XGB model trained on D2 5.2.1.2) with HPO applied across all models. Discrimination remains effectively unchanged compared to D2, with test AUC values tightly clustered around 0.71.

Table 5.4: Predictive performance across model families for D3 (tuned design).

Model	CV AUC	Train AUC	Val AUC	Test AUC	Brier Score	F1 Optimal
Logistic Regression	0.716	0.718	0.717	0.712	0.212	0.459
XGBoost	0.718	0.736	0.719	0.713	0.209	0.462
Random Forest	0.717	0.766	0.718	0.713	0.205	0.462
MLP	0.713	0.743	0.710	0.712	0.207	0.467

5.2.2 Cross design comparison

The best test AUC increased from 0.582 in D1 5.2.1.1 (achieved by MLP) to 0.715 in D2 5.2.1.2 by XGB, representing an improvement of 0.133, demonstrating the benefit of incorporating pooled longitudinal transitions.

Reducing the feature set to the top-20 predictors in D3 maintained this gain, with XGB achieving a test AUC of 0.713, only 0.002 lower than D2, confirming that most of the predictive signal is concentrated in a small subset of variables, and that performance is robust to feature reduction.

Across designs, CV performance also improves and stabilises:

- D1: 0.589 ± 0.072 (LR)

- D2: 0.716 ± 0.007 (XGB)

- D3: 0.718 ± 0.006 (XGB)

The reduction in standard deviation from D1 to D2 and D3 reflects improved stability associated with the pooled multi-wave design and more consistent learning signal across transitions, although differences in data structure across designs should be considered when interpreting cross design variability.

5.2.3 Model comparison

Once temporal structure is introduced (D2 and D3), differences between model families are modest. Test AUC values cluster in the range (0.69–0.72), with no single model class dominating performance. XGB achieved the highest test AUC in both D2 (0.715) and D3 (0.713), though the margin over LR was narrow (≤ 0.006). The fact that the linear LR remained consistently competitive (0.709 in D2 5.2.1.2 and 0.712 in D3 5.2.1.3) suggests that risk signals in the ageing population are primarily linear. MLP slightly underperforms in D2 5.2.1.2 (0.699) but matched other models in D3 5.2.1.3 (0.712), indicating improved performance when the feature space is reduced to the most informative predictors. This suggests that individual health indicators, such as physical pain or medication use count, contribute largely and independently to the risk score, with limited additional benefit from modelling complex non-linear interactions.

Figure 5.2 compares generalisation gaps (Train−Test) AUC across all three designs. The average train-test AUC gap (across all four models) drops from 0.202 in D1 to 0.030 in D2 and 0.028 in D3 – a reduction of over 85%. This demonstrates that the pooled multi-wave design effectively reduces overfitting. D1 exhibits the largest gaps, consistent with weaker signal and higher variance in estimates. In contrast, D2 and D3 show much smaller gaps, indicating improved stability. Tree based models (XGB and RF) exhibit the largest gaps in D1, consistent with the small sample size. In D2 and D3, XGB achieves the smallest gaps (0.021 and 0.023 respectively), while RF shows comparatively larger gaps (0.029 and 0.053). The strong performance of XGB in D3 is expected, as the top-20 feature set was selected using XGB gain importance from D2. Nevertheless, XGB also generalises well in D2, suggesting that its boosting mechanism is effective at learning stable patterns from pooled longitudinal data. In contrast, RF may be more prone to overfitting even with the pooled design, possibly due to its reliance on bagging and random feature selection, which may capture informative predictors less efficiently than boosting.

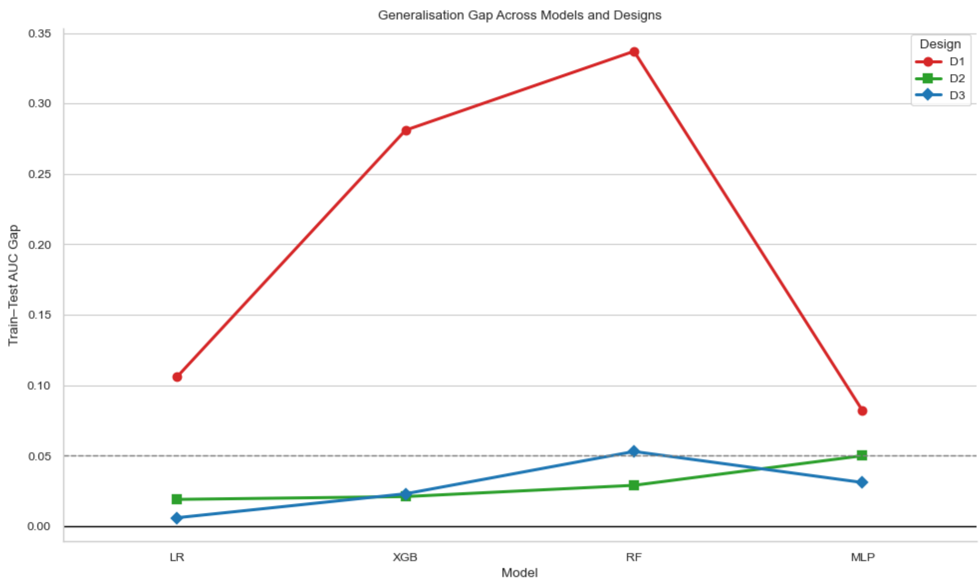

Figure 5.2: Train−Test generalisation gaps across models and designs. 0.05 marks a conventional heuristic threshold for acceptable error in AUC. Gaps below this level indicate that the model generalises well to unseen data.

The figure also highlights that model level differences in generalisation gap are more pronounced in D1 5.2.1.1, whereas D2 5.2.1.2 and D3 5.2.1.3 exhibit convergence across model families, suggesting that data structure plays a larger role than algorithmic choice in controlling overfitting.

5.2.4 Track B: Disease specific

Track B focuses on individual chronic disease outcomes. Each model targets a single condition using the pooled multi-wave cohort structure, participant level splitting, and XGB classifier established earlier in D2 and D3. Twenty four disease specific outcomes were evaluated using two feature strategies: all eligible predictors retained after the 50% missingness filter (for example, 261 features for functional decline and 351 for circulatory disease), and a restricted top-20 set derived from XGB gain importance.

The outcomes span key domains of ageing related health, including cardiovascular conditions, functional decline, mental health, and metabolic disease, ensuring that the models cover a broad range of clinical relevance in the TILDA cohort.

5.2.4.1 Outcome categories

- **Clinical outcomes (ICD10 codes):** Circulatory disease, endocrine and metabolic diseases, eye diseases, and musculoskeletal diseases, derived from WHO `ICD10` codes. These codes were only available comprehensively from Wave 5, with limited coverage in earlier waves, as described in Subsection 5.2.1.1. All source predictor variables that directly encode each condition were excluded to prevent leakage.

- **Functional and symptom outcomes:** Functional decline, falls and balance, chronic pain, hearing loss, and urinary incontinence, from self-reported health variables. These outcomes are available across multiple waves and support pooling across multiple transitions (e.g., W2→W4, W3→W4→W5 and W2→W4).

- **Family history indicators:** Variables recording whether a participant reports that a family member has been diagnosed with a condition, covering diabetes, high cholesterol, hypertension, heart disease, osteoporosis, Alzheimer's disease, cancer, and depression. These are not incident diagnoses of the participant and represent a distinct prediction target: estimating the likelihood of a participant newly reporting a family history item at follow-up. These variables were retained through the manual filtering process described in Section 3.4.

- **Medication and procedure proxies:** Antidepressant and antihypertensive prescription, pain analgesia, cholesterol medication, doctor confirmed hypertension, and cardiac procedure. These are downstream markers of clinical decision making. When used as prediction targets, they capture risk signals linked to eventual treatment rather than raw disease onset. As noted in (Appendix H), incidence rates for medication proxies vary widely, from 1.4% for cardiac procedure to 41.6% for doctor confirmed hypertension diagnosis (though this is based on a very small cohort of $n = 250$).

5.2.4.2 Overall performance

Figure 5.3 plots train AUC against test AUC for all disease specific models, coloured by outcome category. The diagonal represents perfect generalisation (train equals test), while dashed lines at 0.5 on both axes mark chance level performance. Points close to the diagonal indicate stable learned signal, whereas points far above it, with much higher train than test AUC, indicate overfitting. Overall, test AUC values range from approximately 0.45 to 0.72, reflecting substantial variation driven mainly by outcome type and incidence rate rather than model choice.

Functional and symptom outcomes were the most reliable across both feature strategies. Functional decline ($n = 14{,}296$, incidence 22.5%) achieved a test AUC of 0.713 with a small gap between train and test, consistent with the strong and persistent frailty signal identified in the literature. FOF ($n = 18{,}114$, incidence 14.0%) achieved 0.721, the highest test AUC, closely followed by antihypertensive prescription, reflecting the well documented link between functional limitation and falls risk in TILDA. Hearing loss (n = 10,780, incidence 15.1%) reached 0.678, and urinary incontinence (n = 16,208, incidence 8.4%) achieved 0.682, both with gaps below 0.10. These outcomes benefit from consistent measurement across waves, sufficient event counts, and clear physiological pathways, which support more stable longitudinal prediction.

Medication proxies achieved competitive test AUC values. Antihypertensive prescription (n = 19,845, incidence 4.5%) reached 0.718, and antidepressant prescription (n = 4,378, incidence 3.2%) reached 0.702, despite low incidence. Both models showed train AUC above 0.99, indicating overfitting to training specific patterns rather than fully generalisable risk patterns. The test results shows that the models can discriminate between cases and non cases, although part of this performance likely comes from established treatment profiles. For example, a participant already receiving antihypertensives at baseline has a clinical profile strongly associated with future prescription continuation. These models are therefore informative for risk stratification but should not be interpreted as predicting new disease onset.

Clinical `ICD10` outcomes showed moderate performance. Circulatory disease (n = 14,539, incidence 12.2%) and eye diseases (n = 4,396, incidence 10.1%) were the strongest, reaching test AUC values of 0.626 and 0.653, respectively. This aligns with findings that cardiovascular outcomes in TILDA are supported by well established clinical predictors. Musculoskeletal diseases (n = 4,269, incidence 6.5%) and endocrine and metabolic diseases (n = 4,117, incidence 8.8%) performed near chance, with test AUC values of 0.507 and 0.515 under the all features strategy, alongside train AUC values of 1.000 and 0.967. This pattern indicates that the models memorised training labels without learning signal that transfers to new data. This issue is discussed further in Section 5.2.4.4.

Family history models produced the most variable and least reliable results. Using all features, test AUC ranged from 0.513 for cancer (n = 3,971, incidence 13.4%) to 0.665 for high cholesterol (n = 4,312, incidence 12.0%), while training AUC reached 1.000 for several outcomes such as Alzheimer's disease and cancer, indicating severe memorisation. Switching to the top-20 feature set consistently degraded performance, with several models dropping below 0.50. The root cause is the indirect nature of the target: family history variables capture a participant's reported knowledge of a relative's diagnosis, rather than an outcome experienced by the participant. As a result, there is no direct causal pathway linking participant level predictors to these outcomes, limiting the model's ability to learn generalisable patterns. What is learned from the training data therefore doesn't transfer well to unseen data.

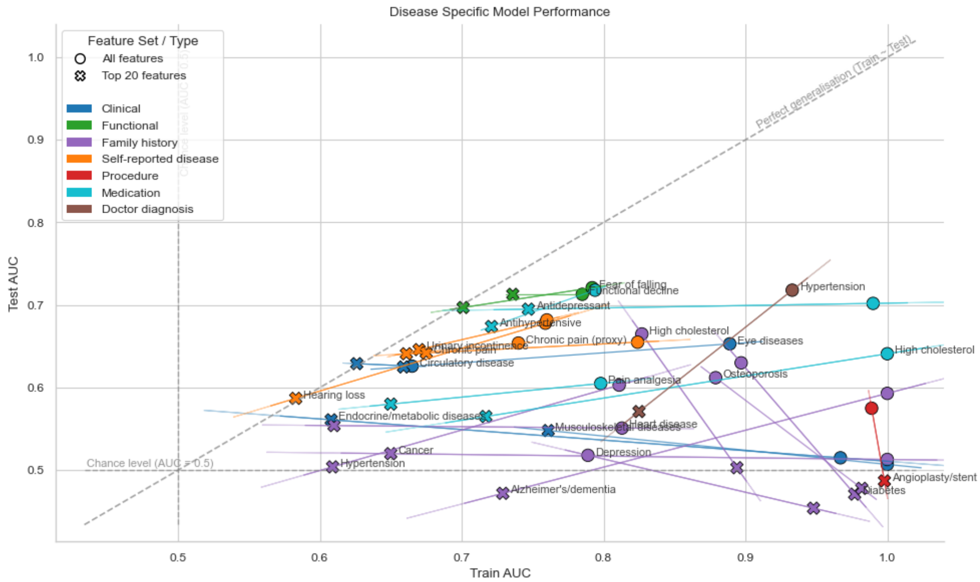

Figure 5.3: Train AUC vs Test AUC for all Track B disease specific models, coloured by outcome category. Circle markers represent all feature models and cross markers represent top-20 feature models, with a line connecting the two feature strategies per disease. The diagonal dashed line marks perfect generalisation (Train ≈ Test). Dashed lines at 0.5 on both axes indicate chance level discrimination. Confidence ellipses illustrate within category spread.

5.2.4.3 Incidence and predictability

Cohort sizes and incidence rates for each outcome are provided in Appendix H. Incidence rate appears to be the most consistent driver of model stability in Track B. Outcomes with incidence above approximately 12%, particularly when measured consistently across waves, tended to achieve test AUC values above 0.65. In contrast, outcomes with incidence below approximately 7% generally produced near chance test AUC and large gaps between train and test regardless of feature strategy.

A model trained on few positive examples cannot learn stable decision boundaries. Class imbalance weighting via `scale_pos_weight` partially compensates, but when positive cases are very sparse, the model can achieve high training scores by memorising those few cases without generalising. This is why Train = 1.000 appears for musculoskeletal diseases, Alzheimer's disease, and cancer in the all features models, none of which exceed test AUC of 0.54 despite high training performance.

Doctor confirmed hypertension is an outlier worth noting. Its incidence rate of 41.6% is the highest in Track B, and it achieved test AUC of 0.718. However, the cohort is very small at $n = 250$, making this estimate less reliable. This result should be interpreted with caution, as a cohort of this size can produce highly variable test scores depending on the specific participants allocated to the test split.

5.2.4.4 Overfitting and feature reduction

For outcomes where the all features strategy produced train AUC substantially above 0.90 alongside near chance test AUC, the top-20 strategy provides a more reliable estimate of generalisable performance, even when absolute performance remains modest. Musculoskeletal diseases moved from train 1.000 / test 0.507 to train 0.761 / test 0.548 under feature reduction. Endocrine and metabolic diseases similarly improved from 0.515 to 0.561. These are not large gains, but they indicate a more calibrated model that has learned from signal rather than noise. In contrast, for outcomes where both strategies produced similar test AUC with small gaps, such as functional decline, FOF, and hearing loss, the feature reduction confirms stable learning rather than addressing overfitting.

For family history targets, feature reduction worsened rather than helped. With only 20 features, the model is further constrained in its ability to identify even proxy signal for family reported diagnoses. This is consistent with the broader point that the target type, rather than the feature count, drives instability in these models.

The overall pattern across Track B is consistent with the preprocessing decisions made in (Chapter 3.4). Variables were excluded during the manual filtering stage if they exceeded the missingness threshold, exhibited constant values, or had a single response category dominating more than approximately 90% of observations. This process improved overall signal quality, but it also reduced the number of available predictors for outcomes that are inherently sparse or indirectly measured. Outcomes with rich, well distributed predictor signals, such as functional decline and FOF, benefited most. Outcomes dependent on rare or indirect signals, such as Alzheimer's family history or cardiac procedure, remained constrained by both the target definition and the reduced predictor space. The results in Figure 5.3 should therefore be interpreted not only as a function of modelling choices, but as a reflection of data availability and the structural properties of each outcome in TILDA.

5.3 Model interpretability and feature importance

Feature importance analysis across all designs reveals three consistent findings. First, predictive signal is concentrated in a small, highly redundant set of health burden indicators, primarily spanning the Physical/Clinical and Functional thematic domains defined in (Chapter 3.3), with other domains adding little additional discriminative information beyond the baseline feature set. Second, the most important predictors shift fundamentally between designs: objective clinical biomarkers dominate the 8 year prediction window in D1, while medication burden and self-rated health dominate the 2 year transitions in D2 and D3. Third, no single domain specific predictor adds statistically significant predicitive value. This remained true within the constraints of the available features and model specification, as confirmed by the add-on analysis. Full feature importance bar plots for D1 and D2 across all four model families are provided in Appendix J. Figure 5.4 displays the add-on Δdelta visualisation.

5.3.1 Design 1: Clinical biomarkers dominate

In D1, the top-20 features selected by `SelectKBest` were dominated by cardiovascular and anthropometric measurements from the wave 1 health centre assessment (see Appendix J.1). Blood pressure variables, covering seated and standing systolic and diastolic readings and their means, appeared prominently across all four model families. Anthropometric frailty markers, specifically hip circumference (0.084 for RF, 0.037 for MLP), waist circumference, and BMI, were consistently ranked alongside trail making test time as a marker of cognitive processing speed. Social engagement through attendance at classes or lectures, health insurance coverage, mother's age at death as a familial longevity signal, socioeconomic group, and physical impairment count contributed additional signal across models.

This pattern aligns with what the literature review chapter identified as the primary drivers of chronic disease and mortality in this cohort. Blood pressure is a well established cardiovascular risk marker [41], anthropometric measures connect to the frailty and obesity pathway in TILDA [33], and trail making test time reflects the cognitive reserve framework discussed in (Chapter 2) [23]. The dominance of these objectively measured variables reflects the availability of a full wave 1 health centre assessment providing objective clinical measurements that were less consistently available in later waves.

Across model families, the ranking order varied modestly. Seated systolic blood pressure ranked first for both XGB (gain 0.072) and RF, which is notable given that these are both tree based models using different importance metrics, suggesting that the feature is consistently identified as important across tree based approaches. By contrast, LR showed a broader distribution placing attending classes or lectures at rank one (0.039), and MLP placed hip circumference first (0.037). Importance values are not directly comparable across model families as each uses a different scoring method, so the cross model consistency in which features appear, rather than their specific scores, is the more meaningful observation here.

5.3.2 Design 2: Medication burden and self-rated health emerge

Shifting from D1 to D2 produced a major change in the importance landscape (see Appendix J.2). Blood pressure variables, which dominated D1, did not appear in the top-20 for any model in D2. Medication burden and self-rated health became the primary drivers instead.

Medication count excluding supplements ranked first in XGB (0.068), RF (0.115), and MLP (0.012) in D2. Polypharmacy indicators, total medication counts including supplements, and extended medication burden measures collectively occupied much of the top 10 across all four model families. This shift reflects the change in prediction target from ICD10 coded hospitalisation over 8 years (D1) to self-reported chronic illness over 2 year transitions (D2). Over shorter windows, medication burden is a closer indicator of health deterioration: a participant already taking multiple medications at baseline is substantially more likely to report chronic illness at the next wave.

Self-rated general health ranked first for LR (0.150) and appeared in the top six for all model families. This predictor is one of the most consistently reported predictors of subsequent morbidity and mortality in older adult cohorts [6], and its prominence here validates the predictive framework. Comparative self-rated health, capturing how participants assess their health relative to peers of the same age, also appeared in the top-20 across LR, XGB and RF models.

Pain related variables contributed substantially in D2. Chronic pain limiting activities, pain severity, and any chronic pain indicator all ranked consistently across model families. Chapter 2.6 identified chronic pain as a marker of underlying functional decline and a strong predictor of subsequent disease burden [16]. Functional disability items, specifically difficulty getting up from a chair after sitting, chair rising steadiness, and aggregate physical limitation counts, rounded out the top predictors. Healthcare utilisation variables, general practitioner visit frequency and outpatient visit frequency, also appeared, reflecting that participants who engage heavily with the health system consistently showed elevated risk of incident chronic illness at the next wave.

5.3.3 Cross model consistency and the D3 feature set

A notable finding across D2 is the high degree of agreement between model families regarding which features matter. The top five features by XGB gain, RF Gini importance, LR absolute coefficient magnitude, and MLP permutation importance show substantial overlap in the highest ranked predictors, with only minor ordering differences. This cross model agreement strengthens confidence that these represent genuine risk signals rather than model specific effects of any single modelling approach.

The D3 and Track B top-20 feature set was derived directly from XGB gain importance in D2 and applied unchanged across all four D3 model families and all Track B disease specific models. Table 5.5 lists this feature set with thematic domain assignments following the domains described in Section 3.4.

Table 5.5: Top-20 features from XGB gain in D2, applied in D3 and Track B.

Feature label	Description	Thematic domain
Reported medications (excl. supps)	Total reported medications excluding supplements	Physical/Clinical
Chronic pain (proxy)	Frequency of pain experienced	Functional/Symptom
Polypharmacy (excl. supps)	Taking five or more medications excluding supplements	Physical/Clinical
Regular medications	Number of regular medications including supplements	Physical/Clinical
Reported medications	Number of reported medications including supplements	Physical/Clinical
Self-rated health	General self-rated health (excellent to poor)	Physical/Clinical
Number of physical limitations	Count of activity limitations from mobility items	Functional
Health limits activities	Health problem limits kind or amount of activities	Functional
Pain limits activities	Chronic pain restricts daily activities	Functional/Symptom
GP visits (12 months)	Number of GP visits in past 12 months	Physical/Clinical
Large muscle ADL/IADL index	Composite of large muscle daily activity items	Functional
Pain severity	Level of pain reported	Functional/Symptom
Flu vaccination	Had a flu vaccination in past 12 months	Lifestyle
Walking/standing steadiness	Steadiness when walking or standing	Functional
Chronic pain	Any chronic pain present	Functional/Symptom
Chair rising steadiness	Steadiness when getting up from a chair	Functional
Hospital visits (12 months)	Number of outpatient visits in past 12 months	Physical/Clinical
Getting up from chair difficulty	Difficulty getting up from a chair after sitting	Functional
Functional difficulty	Difficulty stooping, kneeling, or crouching	Functional
Health vs peers	Self-rated health compared with others of same age	Physical/Clinical

The feature set spans two primary thematic domains: Physical/Clinical (medication counts, self-rated health, healthcare utilisation) and Functional (mobility difficulty, pain, physical limitation), with one Lifestyle item (flu vaccination).

Nutritional and metabolic biomarkers, psychological and cognitive measures, and sociodemographic variables are absent from the top-20.

5.3.4 Predictor add-on analysis

The add-on analysis evaluated the incremental contribution of individual candidate features to the prediction of functional decline using a BASE versus BASE+ ADDON framework. For each candidate, a model was trained on all available predictors except that candidate (BASE), then retrained with it included. The baseline cohort and feature set are determined dynamically for each candidate. Only waves in which both the outcome target and the candidate feature were present were used, meaning cohort size and feature count varied across candidates. For example, when a history of blackout or fainting served as the candidate feature, the analysis included waves 1 to 3, yielding 929 held-out test participants and approximately 570 features entering the model after the missingness filter. When education level served as the candidate, waves 3 to 5 were used, yielding 660 test participants and approximately 464 features. The delta in AUC and (Average Precision (AP)) was estimated with bootstrapped 95% CIs ($B = 2{,}000$ resamples on the held-out test set). Full delta results are shown in (Figure 5.4). The candidates span lifestyle, socioeconomic, cognitive, nutritional, and social domains, including features from thematic domains 3.2 not represented in the top-20 feature set.

Given the large baseline feature sets entering each comparison, marginal gains from a single additional candidate feature are expected to be small. The purpose of the analysis is therefore to assess whether any domain contributes unique predictive information beyond this baseline rather than to achieve statistically significant improvements. No candidate produced a statistically significant improvement across any metric. Every AUC delta CI crossed zero. Selected results, organised by domain cluster, are as follows.

(i) **Functional predictors** (chair rising difficulty, physical limitations, chair rising steadiness, FOF). Among the functional candidates, the highest BASE AUC occurred when getting up from a chair difficulty served as the candidate feature (0.707). Adding it to the BASE produced a near zero marginal gain (CI: $[-0.012, 0.002]$). Physical limitation count (CI: $[-0.008, 0.007]$) and chair rising steadiness (CI: $[-0.006, 0.010]$) showed similarly null deltas. The NOS table in Chapter 2.3 identified gait speed, chair stand performance and physical limitation as primary frailty indicators in TILDA [7]. The small incremental contributions are consistent with strong correlation across functional predictors: pain, steadiness, and mobility difficulty items share their predictive variance, and none appears to contributes substantial independent information beyond the others.

(ii) **Socioeconomic predictors** (education level, household income). Education level produced the largest observed positive delta, with Test AUC moving from 0.711 to 0.719 (CI: $[-0.002, 0.018]$). The direction is consistent with the examined literature on social disadvantage [15], although the interval crosses zero and the effect cannot be confirmed statistically. Total household income produced a smaller positive trend ($[-0.004, 0.010]$). The smaller effect likely reflects alignment with healthcare utilisation and employment status predictors already in the BASE.

(iii) **Psychosocial and mental health predictors** (UCLA loneliness, depressive symptoms, social connectedness). UCLA loneliness produced no detectable effect (CI: $[-0.009, 0.006]$). Depressive symptoms produced a small positive delta of +0.001 (CI: $[-0.008, 0.010]$). Social connectedness showed similarly minimal movement (CI: $[-0.008, 0.007]$). Chapter 2.3 documents bidirectional links between SOC (social isolation, depression, and functional health) in TILDA [34, 36]. The null deltas do not contradict these associations, they indicate that the relevant pathways are already captured within the functional and pain predictors present in the BASE.

(iv) **Physical and behavioural predictors** (grip strength, BMI, cigarettes per day, CAGE alcohol scale, history of blackout or fainting). Grip strength produced a positive delta of +0.004 (CI: $[-0.004, 0.012]$). Chapter 2 identifies grip strength as a primary frailty marker in TILDA [26]; the near zero increment confirms that the functional limitation variables in the BASE already capture this frailty related signal. BMI produced +0.002 (CI: $[-0.007, 0.012]$).

Cigarettes per day trended slightly negative ($[-0.012, 0.002]$), and the CAGE alcohol scale contributed minimally ($[-0.004, 0.008]$). A history of blackout or fainting produced the highest BASE AUC in the study (0.714, CV 0.726$\pm$0.009), indicating a BASE model that is already performing strongly when this variable is withheld. Adding it back yielded no further gain (CI: $[-0.007, 0.007]$), suggesting its signal is captured through correlated cardiovascular and health burden predictors already present in the BASE.

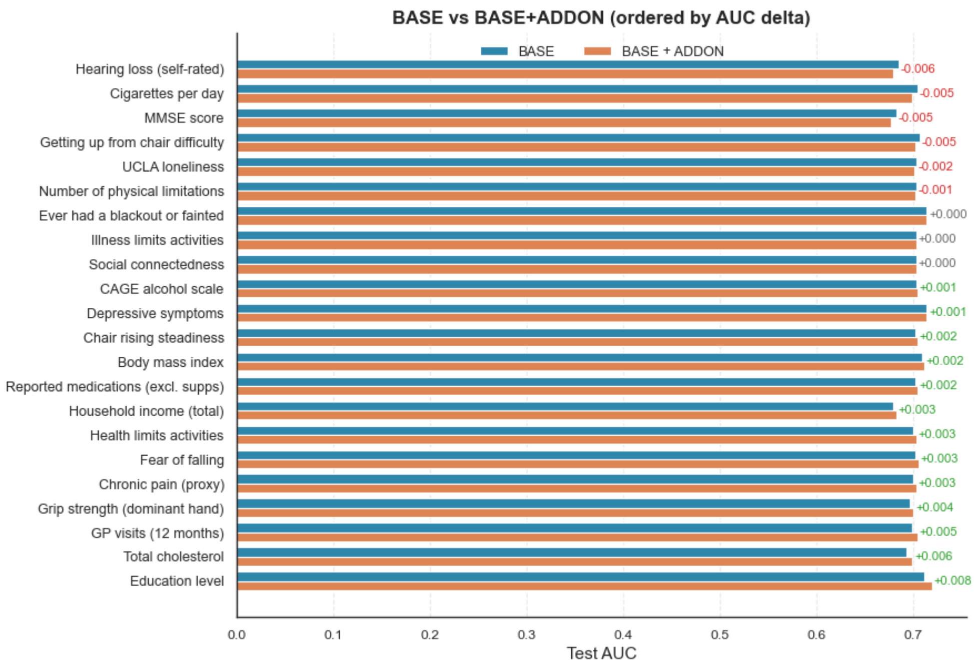

Figure 5.4: Bootstrapped 95% CIs for Test AUC delta (BASE+ADDON minus BASE) across all candidate features (B = 2,000 resamples). Features are sorted in ascending order by the midpoint of the delta interval. All intervals cross zero, indicating no statistically significant incremental contribution from any candidate.

Across all designs and model families, predictive signal across these related health outcomes was concentrated within a highly correlated pattern of indicators associated with accumulated health burden. Medication load, self-rated health, pain frequency and severity, and mobility difficulty collectively capture most of the discriminative signal available in this dataset: social, cognitive, nutritional, and behavioural domains showed weak positive trends but no independent contribution within the context of the available feature set and model specification. This is supported by two convergent findings. Restricting D3 to 20 features produced no reduction in performance, and no add-on candidate improved discrimination significantly. Together, these results suggest that for 2-year transitions in this cohort, the signal structure reflects a highly correlated burden indicators rather than a set of separable domain specific risk factors.

5.4 Health burden correlation structure

The results in this chapter reflect not only modelling choices but also the structure of the underlying data. This section examines pairwise correlations among the key health burden variables used in this study to explain three recurring patterns: why pooled designs outperformed the single wave baseline, why reducing the feature set caused negligible loss in performance, and why no add-on candidate produced statistically significant independent gain.

5.4.1 Shared health burden signal

Figure 5.5 presents pairwise Spearman correlations at Wave 5 (n = 4,978) for six health burden variables used across Track A and Track B: self-reported long term health problem, long term illness, any clinically coded chronic disease (`ICD10_ANY`), chronic pain, self-rated health, and medication use.

The most striking feature of the matrix is the perfect correlation ($r = 1.00$) between self-reported long term health problem and long term illness. Both share identical prevalence (44.2%, n = 2,201 out of 4,977 valid responses) and are empirically indistinguishable within this wave of TILDA. They encode a single construct, and predictive performance for either target reflects the same signal rather than two independent outcomes.

A more significant pattern is the consistently weak alignment between `ICD10_ANY` and all self-reported measures. Correlations range from $r = 0.094$ with chronic pain to $r = 0.245$ with medication use, with long term illness at $r = 0.168$ and self-rated health at $r = -0.174$. These weak associations are not driven

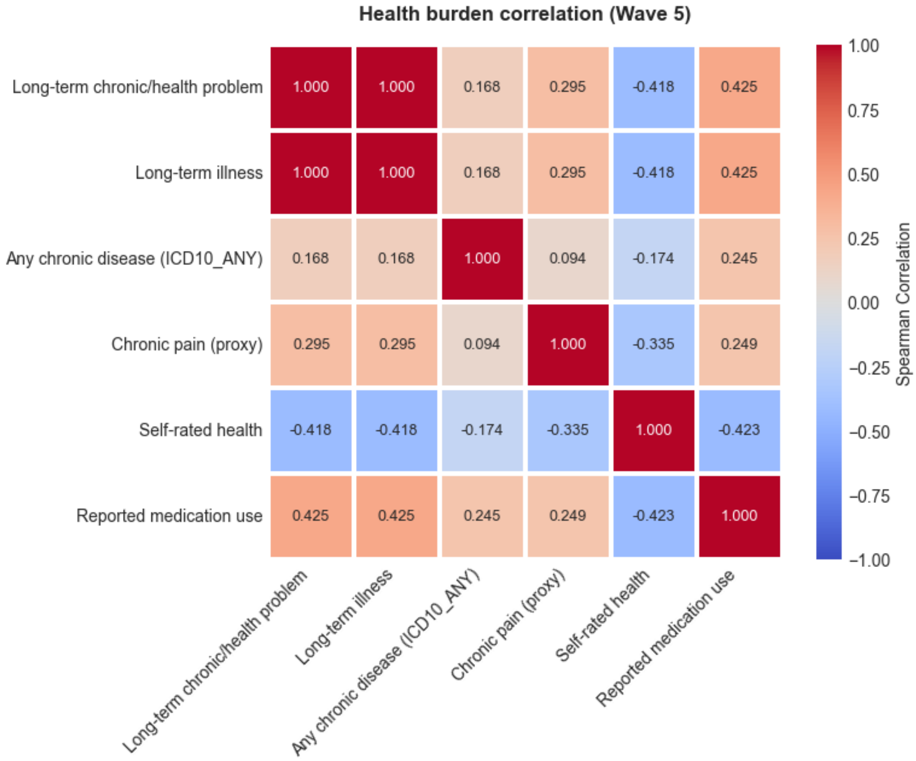

Figure 5.5: Pairwise Spearman correlations among six health burden variables at Wave 5 (n = 4,978). Self-reported long term illness and chronic health problem form an identical construct ($r = 1.00$), while `ICD10_ANY` shows consistently weak associations with all self-reported measures.

by imbalance: `ICD10_ANY` prevalence was 37.6%, comparable to the self-reported outcomes. They reflect a genuine measurement gap, and as confirmed by the Cohen's Kappa analysis in Section 5.1.1 ($\kappa = 0.167$), the agreement between them is poor.

All top-20 variables identified in D2 and D3, including medication burden, self-rated health, chronic pain, mobility limitation, and healthcare utilisation, are survey derived and naturally aligned with the self-reported layer rather than with clinically coded records. The issue is not that these predictors are uninformative, but that they are aligned more closely with subjective and functional health states than with clinically coded disease definitions. This explains why D1, which used `ICD10_ANY` as its outcome, produced substantially weaker and less stable discrimination than D2 and D3.

A third pattern concerns the internal structure of the self-reported measures. Medication use, self-rated health, chronic pain, and long term illness are moderately correlated with pairwise values ranging from $|r| = 0.25$ to $|r| = 0.43$. Self-rated health is inversely associated with all other burden indicators: medication use ($r = -0.423$), chronic pain ($r = -0.335$), and long term illness ($r = -0.418$). These moderate relationships reflect a shared dimension of accumulated health burden while still preserving some variable specific information.

This internal structure directly explains why the add-on analysis in Section 5.3 produced no statistically significant results. Add-on candidates from outside this set, including education level, grip strength, UCLA loneliness, and cognitive function (MMSE), were each tested against a BASE model already containing the core burden indicators. None produced a measurable increment, because the principal signal had already been absorbed by these variables.

Prediction of chronic disease risk in this cohort is therefore governed by a compact and internally correlated set of perceived and functional health measures. Further gains from adding variables of the same type are unlikely. The present analysis used the publicly accessible TILDA release (Waves 1–5), which contains self-reported health, functional, cognitive, nutritional, and behavioural variables, together with `ICD10` coded disease indicators from Wave 2 onwards. Richer data that could distinguish perceived health burden from objective physiological decline, including linked mortality records, primary care records, and repeated biomarker panels, are held under elevated access and were not available for this study. The inclusion of family history indicators in Track B partly reflects this constraint: participant-level onset data for specific conditions were often sparse or absent in the public release, and family reported diagnoses served as an indirect proxy where direct measures were unavailable.

These considerations provide the context for the discussion in Chapter 6, where the implications of the modelling results and directions for future work are considered.

Chapter 6

Conclusion

6.1 Summary

Models x, y, z

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/restrictions: The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to https://tilda.tcd.ie/data/accessing-data/.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by the `Faculty of Health Sciences Research Ethics Committee` at Trinity College Dublin, Ireland. The patients/participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

To use wave6 published on 15/12/2025, IDS and covid-19 datasets. How well our models generalise well compared to international data. The utilisation of the harmonised dataset.

Appendix A

Python code for exporting PMF metadata to CSV

```python
import pandas as pd
import pyreadstat
from pathlib import Path
import json

data_files = {
    "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01_TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
    "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data/0053-02_TILDA_Wave2_PMF_v2.7.sav",
    "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data/0053-04_TILDA_Wave3_PMF_v3.6.sav",
    "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05_TILDA_Wave4_PMF_v4.4.sav",
    "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06_TILDA_Wave5_PMF_v5.5.sav",
    "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data/0053-03_TILDA_Harmonized_v1.2.dta",
}


def load_df_with_var_of_interest(
    var_dir="var_of_interest", data_files=data_files, verbose=False
```

```python
):
    dfs = {}
    var_dir = Path(var_dir)

    for json_name, data_path in data_files.items():
        file = var_dir / json_name
        with open(file, "r") as f:
            vars_list = json.load(f).get("variables_of_interest")
            if verbose:
                print(f"Variables of interest from {file.name}: {vars_list[:10]} ...")

        if data_path.endswith(".sav"):
            df_full, meta = pyreadstat.read_sav(
                data_path, encoding="latin1", apply_value_formats=True
            )
            df, _ = pyreadstat.read_sav(
                data_path,
                usecols=vars_list,
                encoding="latin1",
                apply_value_formats=True,
            )
        else:
            df_full = pd.read_stata(data_path, convert_categoricals=True)
            df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]

        if verbose:
            print(
                f"Data shape (Original): {df_full.shape[0]} rows x {df_full.shape[1]} columns"
            )
            print(f"Data shape (Cleaned): {df.shape[0]} rows x {df.shape[1]} columns")
            print("-----------------------------------------------")

        dfs[file.stem] = df

```

```
    return dfs


if __name__ == "__main__":
    for json_name, path_str in data_files.items():
        path = Path(path_str)
        wave_name = json_name.replace(".json", "")
        if path_str.endswith(".sav"):
            # SPSS
            df, meta = pyreadstat.read_sav(
                path, encoding="latin1", apply_value_formats=True
            )

        else:  # Stata .dta file
            df, meta = pyreadstat.read_dta(path, encoding="latin1")

        pd.DataFrame(
            {"variable": meta.column_names, "label": meta.column_labels}
        ).to_csv(Path("output") / f"{wave_name}_variable_descriptions.csv", index=False)

        print(f"--------------- wave {path} ---------------")
        print(df.shape)
        print(df.columns[:20])
        print(
            "Variable/Feature descriptions, csv file saved successfully -> dir=output/"
        )
        df.to_csv((Path("output") / f"{wave_name}.csv"), index=False)

    dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Scenario: Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	**809**	**482**	**639**	**592**	**577**	**453**

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Appendix C

First Stage: Scenario Variable Mapping

```python
# Sentinel values (numeric and string)
# -1 and -1.0 is sentinel but will be used in the clean pipeline
standardised_sentinels = {'-1', '-1.0', -1, -1.0} #  set -> duplicates removed -> counts correct -> single value detection is zero
numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
                     95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}" for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "refuse", 'none', 'no response', 'nan', 'not answered', 'no answer',
     'not applicable','no consent','dont know',"don't know","didn't answer",'prefer not to say', 'task not done', 'insufficient sample', 'unable to record',
     '__', '*']
blanks = ['', ' ', '\t', '\n']

SCENARIOS = [
    "Empty / blank values",
    "Numeric with sentinel",
    "Clean numeric",
    "Censored / bracketed numeric",
    "Binary categorical",
```

```python
    "Multi class categorical / mixed",
    "Single value categorical",
    "Special symbol / mixed",
    "Unknown / complex mixed"
]


# flatten all dtypes across the waves to float and category
def flatten_column_types(df):
    df_flat = df.copy()

    for col in df_flat.columns:
        if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric columns
            df_flat[col] = df_flat[col].astype(float)  # Convert to float
        else:
            df_flat[col] = (  # Convert to string, strip whitespace, lowercase, then category
                df_flat[col]
                .astype(str)
                .str.strip()
                .str.lower()
                .astype('category')
            )

    return df_flat

def detect_column_scenario_mapping(df, sample_size=10):
    summary = []

    # Regex patterns
    special_symbols_pattern = r'[^A-Za-z0-9\s\.,\-+&/()]' # special symbols excluding common numeric formatting
    bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[<>+-]"  # patterns like "10-20", "30/40", ">=50"
    sentinel_set = set(x.lower() for x in string_sentinels) # Lowercase string sentinels for comparison

    for col in df.columns:
```

```
        col_data = df[col].copy()
        dtype = str(col_data.dtype) # get data type as string

        # Convert to string for sentinel / blank detection
        str_vals = col_data.astype(str).str.lower().str.strip() # lowercase and strip whitespace
        clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(str_vals.isin(blanks))] # remove sentinels and blanks

        # Remove sentinels and blanks for scenario detection only
        vals_no_sentinel = clean_vals[~clean_vals.astype(str).str.lower().isin(sentinel_set)
                                      & ~clean_vals.astype(str).str.lower().isin(blanks)].unique()
        num_unique_no_sentinel = len(vals_no_sentinel)

        # Check if any sentinel values are present
        sentinel_present = str_vals.isin(sentinel_set).any()

        # Detection flags
        empty_detected = str_vals.isin(blanks).any() or len(col_data) == 0

        # Numeric detection: coerce to numeric, NaNs for non-numeric
        numeric_vals = pd.to_numeric(clean_vals, errors='coerce') # coerce non-numeric to NaN
        numeric_like = numeric_vals.notna().all() and len(clean_vals) > 0 # all values are numeric-like after cleaning

        sentinel_detected = str_vals[~str_vals.isin(standardised_sentinels)].isin(sentinel_set).any() # or col_data.eq(-1.0).any() # check for sentinel presence
        censored_detected = clean_vals.astype(str).str.contains(bracketed_numeric_pattern).any() # check for bracketed numeric patterns

        # Binary categorical detection (exactly 2 unique values after cleaning)
```

```
        binary_categorical_detected = False
        if dtype == 'category':
            if num_unique_no_sentinel == 2:
                binary_categorical_detected = True
            elif num_unique_no_sentinel == 1 and sentinel_present:
                binary_categorical_detected = True

        # Multi-class categorical (3+ unique values, object/category)
        multi_class_categorical_detected = num_unique_no_sentinel > 2 and dtype == 'category'

        # Single value categorical
        single_value_detected = num_unique_no_sentinel == 1 and not binary_categorical_detected

        # Special symbols detection (ignore numeric formatting)
        symbol_detected = clean_vals.astype(str).str.contains(special_symbols_pattern).any()

        # Mixed types detection
        mixed_detected = len(set(type(v) for v in col_data)) > 1  # more than one data type present

      # Determine scenario in order of pipeline
        if empty_detected:
            scenario = SCENARIOS[0]
            handling = "Replace numeric/categorical empty/blank with -1.0"
        elif numeric_like and sentinel_detected:
            scenario = SCENARIOS[1]
            handling = "Replace numeric sentinel values with -1.0"
        elif numeric_like and not sentinel_detected:
            scenario = SCENARIOS[2]
            handling = "Keep as numeric"
        elif censored_detected:
            scenario = SCENARIOS[3]
            handling = "Convert numeric ranges to midpoints or keep as categorical"
```

```
        elif binary_categorical_detected:
            scenario = SCENARIOS[4]
            handling = "Encode as 0/1 or keep as category"
        elif multi_class_categorical_detected:
            scenario = SCENARIOS[5]
            handling = "One-hot encode, label encode, or keep as category"
        elif single_value_detected:
            scenario = SCENARIOS[6]
            handling = "Keep as category / constant column"
        elif symbol_detected:
            scenario = SCENARIOS[7]
            handling = "Inspect and preprocess based on context"
        else:
            scenario = SCENARIOS[8]
            handling = "Inspect manually / complex mixed types"

        sample_values = list(col_data.unique()[:sample_size]) # sample unique values

        summary.append({
            'Variable': col,
            'DataType': dtype,
            'UniqueValues': num_unique_no_sentinel,
            'Scenario': scenario,
            'SuggestedHandling': handling,
            'SampleValues': sample_values
        })

    return pd.DataFrame(summary)
```

Appendix D

Step 6: Scenario ‘Multi class categorical’ Cleaning Function

```
def categorical_transform_S6(df): # Step6; Scenario -> Multi class categorical
    df_cleaned = df.copy()
    sent_set = set([str(x).lower() for x in string_sentinels]) | set([str(x).lower() for x in standardised_sentinels])
    numeric_sentinel_set = set(float(x) for x in numeric_sentinels)

    # Helper normaliser
    def _norm(tok):
        if tok is None:
            return None
        s = str(tok).strip().lower()
        if s in blanks:
            return ''
        return s

    # Words -> number map
    word_num_map = {
        'zero':0.0,'one':1.0,'two':2.0,'three':3.0,'four':4.0,'five':5.0,'six':6.0,'seven':7.0,'eight':8.0,'nine':9.0,
        'ten':10.0,'eleven':11.0,'twelve':12.0,'thirteen':13.0,'fourteen':14.0,'fifteen':15.0,'twenty':20.0,
```

```python
        'thirty':30.0,'forty':40.0,'fifty':50.0,'sixty':60.0,'seventy':70.0,'eighty':80.0,'ninety':90.0,
        'a few':2.0,'a couple':2.0,'few':2.0,'once':1.0,'twice':2.0,'several':3.0,'couple':2.0
    }
    def words_to_num(token):
        if token is None:
            return None
        token = token.strip().lower()
        if token in word_num_map:
            return word_num_map[token]
        try:
            return float(token)
        except Exception:
            return None

    # Regexes for parsing
    RE_TIME = re.compile(r'^\s*(\d{1,2}):(\d{2})\s*$')
    RE_PERCENT = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*%\s*$')
    RE_DIRECT_NUM = re.compile(r'^\s*-?\d+(?:\.\d+)?\s*$')
    RE_RANGE = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*[---/]\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_RANGE_TO = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*(?:to|TO)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_PLUS = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*\+\s*$')
    RE_GT = re.compile(r'^\s*(?:>=|>)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_LT_ANCHOR = re.compile(r'^(?:<|<=)\s*-?\d+(?:\.\d+)?')
    RE_NUM_FIND = re.compile(r'-?\d+(?:\.\d+)?')
    RE_WORD_TOK = re.compile(r"[a-z']+(?:\s[a-z']+)?")
    # RE_SURVEY_TIME = re.compile(r'^\s*(?:[a-z]+\s+)?(\d+(?:\.\d+)?)\s*(?:month|week|year)s?.*$', re.IGNORECASE)

    def norm_return(v):
        try:
            vf = float(v)
        except Exception:
            return None
        if vf in numeric_sentinel_set:
            return 'SENTINEL'
        return vf

```

```python
    def try_parse_token(x):
        """Return float, 'SENTINEL', or None for unknown token x"""
        if x is None:
            return None

        # preserve numeric pre transformed values
        if isinstance(x, (int, float)):
            if float(x) in numeric_sentinel_set:
                return 'SENTINEL'
            return float(x)

        s = _norm(x)
        # Extending sentinel detection for categorical parsing scenario
        if s == '':
            return 'SENTINEL'
        if 'not applicable' in s or 'does not have' in s or "don't receive" in s or "don\x92t receive" in s or 'unknown' in s:
            return 'SENTINEL'
        if s in sent_set or re.fullmatch(r'[^0-9a-z]+', s):
            return 'SENTINEL'
        if s in ('other', 'other specify') or s.startswith('other (') or s.endswith('(specify)'):
            return 'SENTINEL'

        s0 = s.replace(',', '').replace('$', '').replace(EUR, '').replace('#', '')

        s0 = re.sub(r'(\d+)-', r'\1 -', s0)
        s0 = re.sub(r'-(\d+)', r'- \1', s0)

        RE_PREFIX_NUM = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\.\s*.+$')

        m = RE_PREFIX_NUM.match(s0)
        if m:
            # If the format is 'NUM. anything', return the number as the category value
            return norm_return(float(m.group(1)))
```

```python

        # time
        m = RE_TIME.match(s0)
        if m:
            h, mi = float(m.group(1)), float(m.group(2))
            return norm_return(h * 60.0 + mi)

        # percent
        m = RE_PERCENT.match(s0)
        if m:
            return norm_return(float(m.group(1)))

        # direct numeric
        if RE_DIRECT_NUM.match(s0):
            return norm_return(s0)

        # ranges
        m = RE_RANGE.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass
        m = RE_RANGE_TO.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # plus / greater than
        m = RE_PLUS.match(s0) or RE_GT.match(s0)
        if m:
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # less than
        if RE_LT_ANCHOR.search(s0) or 'less than' in s0:
```

```
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # or less / more
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*less', s0)
        if m:
            return norm_return(m.group(1))
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*more', s0)
        if m:
            return norm_return(m.group(1))

        # "more than X but less than Y", "over X but less than Y"
        m = re.search(r'(?:more than|over)\s*e?(-?\d+(?:\.\d+)?)\s*(?:but\s*)?(?:less than|under)\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # "between X and Y" / "between X - Y"
        m = re.search(r'between\s*e?(-?\d+(?:\.\d+)?)\s*(?:and|[---/])\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # survey time based
        time_map_weeks = {
            'one week': 1.0,
            'two weeks': 2.0,
            'a month (4 weeks)': 4.0,
            'a month/4 weeks': 4.0,
            'three months (13 weeks)': 13.0,
            'three months/13 weeks': 13.0,
```

```
            'six months (26 weeks)': 26.0,
            'six months/26 weeks': 26.0,
            'one year (12 months/52 weeks)': 52.0,
            'one year/12 months/52 weeks': 52.0
        }
        if s0 in time_map_weeks:
            return norm_return(time_map_weeks[s0])

        # If a category is '3 months or more', try to extract the number and use upper bound
        if 'months or more' in s0 and RE_NUM_FIND.search(s0):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                months = float(nums[0])
                return norm_return(months * 4.33) # Convert months to weeks (4.33 weeks per month)

        # Frequency calendar based
        if any(tok in s0 for tok in ('day','week','month','year','daily','never','almost every day','once','twice','thrice','three','several','couple')):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                val = (float(nums[0]) + float(nums[1])) / 2.0 if len(nums) >= 2 else float(nums[0])
            else:
                if 'once or twice' in s0:
                    val = 1.5
                else:
                    val = None
                    for w in RE_WORD_TOK.findall(s0):
                        wn = words_to_num(w)
                        if wn is not None:
                            val = wn
                            break
            if val is not None:
                if 'day' in s0:
                    if 'week' in s0:
                        return norm_return(val * 4.33)
                    if 'month' in s0:
                        return norm_return(val)
```

```python
                    if 'daily' in s0 or 'every day' in s0 or 'almost every day' in s0:
                        return norm_return(val * 30.0)
                    return norm_return(val * 4.33)
                if 'week' in s0:
                    return norm_return(val * 4.33)
                if 'month' in s0:
                    return norm_return(val)
                if 'year' in s0:
                    return norm_return(val / 12.0)
                return norm_return(val)


        likert_map = {
            # Likert / ordinals (expanded with quintile/lowest/highest detection)
            'strongly disagree': 0.0, 'disagree': 1.0, 'neither agree nor disagree': 2.0, 'neither': 2.0,
            'agree': 3.0, 'strongly agree': 4.0,
            'very poor': 0.0, 'poor': 1.0, 'fair': 2.0, 'good': 3.0, 'very good': 4.0, 'excellent': 5.0,
            'none': 0.0, 'not at all': 0.0, 'hardly ever': 0.0,
            'a little': 1.0, 'some': 2.0, 'a lot': 3.0, 'alot': 3.0, 'often': 3.0,
            'yes': 1.0, 'no': 0.0, 'sometimes': 2.0, 'rarely': 1.0, 'most or all of the time': 4.0,
            'pass': 2.0, 'fail': 1.0, '100': 100.0,
            # Problem severity mappings
            'no problem': 1.0,'minor problem': 2.0,'moderate problem': 3.0,'major problem': 4.0,
            # Concern level mappings
            'not at all concerned': 1.0,'somewhat concerned': 2.0,'fairly concerned': 3.0,'very concerned': 4.0,
            # Survey frequency mappings
            'daily / almost daily': 7.0, 'once a week or more': 6.0,'twice a month or more': 5.0,'about once a month': 4.0,
            'every few months': 3.0, 'about once or twice a year': 2.0,'less than once a year': 1.0,'never': 0.0,
            'daily': 7.0,'every day': 7.0,'almost every day': 7.0,'most or all of the time': 7.0,
```

```
            'every week': 6.0,'once a week': 6.0,'several times per week': 6.0,
            'twice a month': 5.0,'two a month': 5.0,'several times per month': 5.0,'once or twice a month': 5.0,
            'once a month': 4.0, 'once or twice a year': 2.0,'less than once a month': 1.0,
        }

        if s0 in likert_map:
            return float(likert_map[s0])

        # Quintile / ordinal like tokens: 'lowest', '2nd quintile', 'highest'
        if 'quintile' in s0 or 'lowest' in s0 or 'highest' in s0 or 'middle' in s0 or 'median' in s0:
                if 'lowest' in s0:
                    return norm_return(1.0)
                if 'highest' in s0:
                    return norm_return(5.0)
                if 'middle' in s0 or 'median' in s0:
                    return norm_return(3.0)
                mq = re.search(r'(\d+)(?:st|nd|rd|th)?\s+quintile', s0)
                if mq:
                    return norm_return(float(mq.group(1)))
                nums = RE_NUM_FIND.findall(s0)
                if nums:
                    return norm_return(nums[0])

        if 'adjacent' in s0: # Adjacent boxes: keep semantic average if present
            nums = RE_NUM_FIND.findall(s0)
            if len(nums) >= 2:
                return norm_return((float(nums[0]) + float(nums[1])) / 2.0)
            elif nums:
                return norm_return(nums[0])
            return None

        # Fallback: first number
```

```
        nums = RE_NUM_FIND.findall(s0)
        if nums:
            return norm_return(nums[0])

        return None

    def category_sort_key(u):
        us = _norm(u) or ''
        parsed = try_parse_token(us)
        if isinstance(parsed, (int, float)):
            return (0, float(parsed), us)
        if us == 'no' or us.startswith('no ') or us.startswith('not '):
            return (1, 0.0, us)
        if 'yes' in us:
            if 'both' in us or 'two' in us:
                return (1, 2.0, us)
            if 'one' in us:
                return (1, 1.0, us)
            return (1, 1.0, us)
        amount_keywords = ['none', 'not at all', 'hardly ever', 'a little', 'some', 'a lot', 'alot', 'often', 'sometimes', 'rarely']
        for i, key in enumerate(amount_keywords):
            if key in us:
                return (2, float(i), us)
        if 'adjacent boxes' in us or 'adjacent numbers' in us:
            nums = RE_NUM_FIND.findall(us)
            if len(nums) >= 2:
                a, b = float(nums[0]), float(nums[1])
                return (10, (a + b) / 2.0, us)
            elif nums:
                return (10, float(nums[0]), us)
            return (10, 0, us)
        return (99, 0, us)

    # Operate only on categorical columns
    cat_cols = df_cleaned.select_dtypes(include='category').columns.tolist()
    for col in cat_cols:
```

```
        uniques = list(dict.fromkeys(df_cleaned[col].tolist())) # Preserve original tokens with types (do NOT cast to str here)
        normalised_pairs = [(u, _norm(u)) for u in uniques] # Build normalised pairs keeping original token too
        # Keep tokens that are non sentinel and parseable (strings or numbers)
        non_sentinel = [u for u, nu in normalised_pairs if (nu and nu not in sent_set and nu not in blanks and try_parse_token(nu) != 'SENTINEL')]

        if not non_sentinel:
            continue

        # Computing parse results keyed by normalized string (but keep list of originals too)
        parse_results = { _norm(u): try_parse_token(_norm(u)) for u in non_sentinel }

        numeric_count = sum(1 for v in parse_results.values() if isinstance(v, (int, float)))
        sentinel_parsed_count = sum(1 for v in parse_results.values() if v == 'SENTINEL')
        all_parse_numeric = (numeric_count + sentinel_parsed_count == len(parse_results))

        if all_parse_numeric:
            # Map token string -> numeric (SENTINEL -> -1.0)
            code_map = { k: (float(v) if isinstance(v, (int, float)) else -1.0) for k, v in parse_results.items() }

            def map_to_numeric(x):
                # Preserve the already numeric cells (except numeric sentinels)
                if isinstance(x, (int, float)) and not pd.isna(x):
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
```

```
                    return -1.0
                xs = _norm(x)
                if not xs or xs in sent_set or xs in blanks or re
                    .fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0
                return float(code_map.get(xs, -1.0))

            df_cleaned[col] = df_cleaned[col].apply(
                map_to_numeric).astype(float)

        else:
            first_seen_norm = []
            seen = set()
            for u in non_sentinel:
                nu = _norm(u)
                if nu not in seen:
                    first_seen_norm.append(nu)
                    seen.add(nu)

            parse_results = {nu: try_parse_token(nu) for nu in
                first_seen_norm}
            max_reserved_val = -1.0
            for nu, val in parse_results.items():
                if isinstance(val, (int, float)):
                    max_reserved_val = max(max_reserved_val, val)
                        # Assuming any parsed float is a reserved
                        ID if it came from the map
            start_i = int(max_reserved_val) + 1
            if start_i == 0: # Ensure we never start at 0 if 0 is
                a reserved category i.e for ('no')
                start_i = 1
            string_tokens = [nu for nu in first_seen_norm if not
                isinstance(parse_results.get(nu), (int, float))]
            sorted_strings = sorted(string_tokens, key=
                category_sort_key)

            code_map_strings = { norm_tok: i for i, norm_tok in
                enumerate(sorted_strings, start=start_i) }

            def map_mixed(x):
```

```python
                if isinstance(x, (int, float)) and not pd.isna(x):# Preserve real numeric cell values
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
                    return -1.0
                xs = _norm(x)
                if xs == '' or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0

                parsed_val = try_parse_token(xs)
                if parsed_val == 'SENTINEL':
                    return -1.0
                if isinstance(parsed_val, (int, float)):# Captures 'no' (0.0), 'yes' (1.0), and other hardcoded likert values.
                    return float(parsed_val)
                if xs in code_map_strings:
                    return float(code_map_strings[xs])
                return -1.0

            df_cleaned[col] = df_cleaned[col].apply(map_mixed).astype(float)

    # Final sweep on numeric columns
    for c in df_cleaned.select_dtypes(include='float').columns:
        df_cleaned[c] = df_cleaned[c].fillna(-1.0)
        df_cleaned[c] = df_cleaned[c].replace(list(numeric_sentinel_set), -1.0)

    return df_cleaned
```

Appendix E

Second Stage: Deterministic *six step* Cleaning Pipeline

```
# Cleaning pipeline to all waves
dfs_cleaned = {}

# flattened dtype df
dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
    items()} # flatten dtypes

# Original df
display(scenario_count_table(dfs_flat).style.set_caption("
    Original␣Table␣-␣pre␣cleaning")) # display original scenario
    count table

# Step 1: Clean empty/missing values
dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
    dfs_flat.items()} # apply step 1 cleaning
display(scenario_count_table(dfs_step1).style.hide(axis="index").
    set_caption("Step␣1:␣Empty␣/␣Blank␣-␣post␣cleaning"))

# Step 2: Clean sentinel values
dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
    items()} # apply step 2 cleaning
display(scenario_count_table(dfs_step2).style.hide(axis="index").
    set_caption("Step␣2:␣Sentinel␣-␣post␣cleaning"))

```

```
# Step 3: Numerical Transformation
dfs_step3 = {wave: numerical_transform(df) for wave, df in dfs_step2.items()} # apply step 3 cleaning
display(scenario_count_table(dfs_step3).style.hide(axis="index").set_caption("Step 3: Numeric Transformation - post cleaning"))

# Step 4: Categorical Transformation for censored scenario
dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in dfs_step3.items()} # apply step 4 cleaning
display(scenario_count_table(dfs_step4).style.hide(axis="index").set_caption("Step 4: Category Transformation S4 - post cleaning"))

# Step 5: Categorical Transformation for binary values
dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in dfs_step4.items()} # apply step 5 cleaning
display(scenario_count_table(dfs_step5).style.hide(axis="index").set_caption("Step 5: Category Transformation S5 - post cleaning"))

# Step 6: Categorical Transformation for multi class/mixed values
dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in dfs_step5.items()} # apply step 6 cleaning
display(scenario_count_table(dfs_step6).style.hide(axis="index").set_caption("Step 6: Category Transformation S6 - post cleaning"))

dfs_cleaned = dfs_step6
```

Appendix F

MD Dictionary: Scenario Grouping and Transformation Function

```python
def show_grouped_vars_for_scenarios(dfs_original, dfs_transformed, scenarios_to_show): # Function to show grouped variables per scenario across waves -> markdown format
    """
    Used to inspect the scenario variables/features that need handling for transformation/cleaning.
    A pre and post cleaning helper function to identify variables that need specific handling and to
    Inspect actual changes for a scenario from previous cleaning step to current step.
    """

    print("# Category Transformation Mapping")
    # print(f"\n## Scenarios: {scenarios_to_show}") # Markdown mapping dictionary header
    for wave in dfs_original.keys():
        print(f"\n### ==================== {wave.upper()} ====================")

        mapping_df = detect_column_scenario_mapping(dfs_transformed[wave]) # Get scenario mapping from transformed/cleaned to see overall changes for all variables
```

```
        # mapping_df = detect_column_scenario_mapping(dfs_original[wave]) # Get scenario mapping from original to see untransformed values

        for scenario in scenarios_to_show:
            vars_in_scenario = mapping_df[mapping_df['Scenario'] == scenario]['Variable'].tolist() # Variables in scenario
            print(f"\n--- Scenario {scenario}: {len(vars_in_scenario)} ---") # Scenario header for pipeline git commit
            if not vars_in_scenario:
                print("\nNo variables found.")
                continue

            # Dictionary: frozen set(values) -> list of variables
            groups = {}

            for var in vars_in_scenario:
                unique_vals = list(dict.fromkeys(dfs_original[wave][var].dropna()))  # Preserve first seen order from ORIGINAL
                key = tuple(unique_vals)  # Use ordered tuple as key

                if key not in groups: # Initialise list if key not present
                    groups[key] = []
                groups[key].append(var) # Append variable to the group

            # Print grouped variables in markdown format friendly for Notion -> mapping dictionary
            for value_set, var_list in groups.items():
                print("\n*Variables:*","`" + ", ".join(var_list) + "`  ") # \n
                print(f"*Pre transformed values:* {list(value_set)}  ")

                # Pull transformed values from dfs_cleaned / specific step using preserved order
```

```
                try:
                    df_before = dfs_original[wave][var_list[0]]
                    df_after = dfs_transformed[wave][var_list[0]]

                    # Build transformed list in same order as original values
                    transformed_vals = []
                    for orig_val in value_set: # Iterate through original values in order
                            mask = df_before == orig_val
                            if mask.any():
                                trans_vals = df_after[mask].unique().tolist() # Get unique transformed values
                                if trans_vals:
                                    transformed_vals.append(trans_vals[0]) # Append first transformed value
                except Exception:
                    transformed_vals = []
                print(f"*Transformed numeric:* `{transformed_vals}`")


scenarios_to_show = SCENARIOS
# scenarios_to_show = ["Clean numeric"]
# scenarios_to_show = ["Censored / bracketed numeric"]
# scenarios_to_show = ["Binary categorical"]
# scenarios_to_show = ["Multi class categorical / mixed"]

show_grouped_vars_for_scenarios(dfs, dfs_cleaned, scenarios_to_show) # prev step == comparison to current step else current step transformed/cleaned for ready .md dictionary
```

Appendix G

MD Dictionary: Scenario Grouping and Transformation Audit Outputs

- `sex`: Binary biological sex indicator (e.g., `Male/Female`, `Female/Male`, prefixed encodings).
- `ph001`: Self-rated general health status measured on an ordinal scale. Survey question: *"Now I would like to ask you some questions about your health. Would you say your health is. . . "*. Responses range from *Poor* to *Excellent*, with additional 'no response' categories.
- `ph009`: Comparative self-rated health relative to people of the same age. Survey question: *"In general, compared to other people your age, would you say your health is. . . "*.
- `bh006`: Frequency based lifestyle variable combining numeric values, bounded ranges (e.g., `20-39`), and textual descriptors (e.g., `Less than 10`, `30+`). Survey question: *"How many cigarettes do/did you smoke on average per day?"*.

G. MD DICTIONARY: SCENARIO GROUPING AND TRANSFORMATION AUDIT OUTPUTS

```
%-----------------------------------------------------------------↲
↪  ------%
%-----------------------MARKDOWN_MAPPING_DICTIONARY----------------↲
↪  ------%
%-----------------------------------------------------------------↲
↪  ------%

# Category Transformation Mapping

### =================== WAVE1 ===================

--- Scenario Empty / blank values: 0 ---

No variables found.

--- Scenario Numeric with sentinel: 0 ---

No variables found.

--- Scenario Clean numeric: 809 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Male', 'Female']
*Transformed numeric:* '[1.0, 0.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Fair', 'Good', 'Very good', 'Excellent',
↪  'Poor', "Don't know"]
*Transformed numeric:* '[2.0, 3.0, 4.0, 5.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* ['20-39', 10.0, -1.0, 15.0, 5.0, '60+',
↪  "Don't Know", 2.0, 8.0, 3.0, '40-59', 4.0, 7.0, 1.0, 6.0, 14.0,
↪  12.0, 16.0, 18.0, 13.0, 9.0, 0.0, 19.0, 11.0, 17.0]
```

```
*Transformed numeric:* '[29.5, 10.0, -1.0, 15.0, 5.0, 60.0, -1.0,
↪  2.0, 8.0, 3.0, 49.5, 4.0, 7.0, 1.0, 6.0, 14.0, 12.0, 16.0, 18.0,
↪  13.0, 9.0, 0.0, 19.0, 11.0, 17.0]'

--- Scenario Censored / bracketed numeric: 0 ---

No variables found.

--- Scenario Binary categorical: 0 ---

No variables found.

--- Scenario Multi class categorical / mixed: 0 ---

No variables found.

--- Scenario Single value categorical: 0 ---

No variables found.

--- Scenario Special symbol / mixed: 0 ---

No variables found.

--- Scenario Unknown / complex mixed: 0 ---

No variables found.

### =================== WAVE2 ===================

...

--- Scenario Clean numeric: 482 ---

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
```

```
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', '20 - 29', 'Less than
↪  10', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 14.5, 24.5, 10.0, 30.0, -1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Very good', 'Good', 'Fair', 'Excellent',
↪  'Poor']
*Transformed numeric:* '[4.0, 3.0, 2.0, 5.0, 1.0]'

...

### ==================== WAVE3 ====================

...

--- Scenario Clean numeric: 639 ---

*Variables:* 'sex, gd002'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001, ph009'
*Pre transformed values:* ['Very Good', 'Good', 'Excellent', 'Fair',
↪  'Poor', 'DK']
*Transformed numeric:* '[4.0, 3.0, 5.0, 2.0, 1.0, -1.0]'

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, '10 - 19', 'Less than 10', '20 -
↪  29', '30+', -98.0, -99.0]
*Transformed numeric:* '[-1.0, 14.5, 10.0, 24.5, 30.0, -1.0, -1.0]'

...
```

```
### =================== WAVE4 ===================

...

--- Scenario Clean numeric: 592 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'

*Variables:* 'ph001'
*Pre transformed values:* ['Good', 'Fair', 'Poor', 'Excellent', 'Very
↪  Good', 'RF', 'DK']
*Transformed numeric:* '[3.0, 2.0, 1.0, 5.0, 4.0, -1.0, -1.0]'

...

### =================== WAVE5 ===================

...

--- Scenario Clean numeric: 577 ---

*Variables:* 'bh006'
*Pre transformed values:* [-1.0, 'Less than 10', '20 - 29', '10 -
↪  19', '30+', "Don't know"]
*Transformed numeric:* '[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]'

*Variables:* 'gd002, sex'
*Pre transformed values:* ['Female', 'Male']
*Transformed numeric:* '[0.0, 1.0]'
```

```
*Variables:* 'ph001'
*Pre transformed values:* ['Very Good', 'Fair', 'Excellent', 'Good',
↪  'Poor']
*Transformed numeric:* '[4.0, 2.0, 5.0, 3.0, 1.0]'

...
```

Appendix H

Cohort sample sizes by design for Track B

Table H.1: 24-incident rates by disease specific for Track B, 10-year prediction window

Disease	Type	Obs. n	Inc. n	Inc. rate %
Circulatory disease	Clinical	14539	1774	12.2
Endocrine/metabolic diseases	Clinical	4117	362	8.8
Eye diseases	Clinical	4396	444	10.1
Musculoskeletal diseases	Clinical	4269	277	6.5
Functional decline	Functional	14296	3217	22.5
Fear of falling	Functional	18114	2536	14.0
Diabetes	Family history	4307	276	6.4
High cholesterol	Family history	4312	517	12.0
Hypertension	Family history	3590	679	18.9
Heart disease	Family history	2770	598	21.6
Osteoporosis	Family history	4779	330	6.9
Alzheimer's/dementia	Family history	4827	309	6.4
Cancer	Family history	3971	532	13.4
Depression	Family history	4760	343	7.2
Pain analgesia	Medication	2152	775	36.0
High cholesterol	Medication	733	181	24.7
Antidepressant	Medication	4378	140	3.2
Antihypertensive	Medication	19845	893	4.5
Chronic pain (proxy)	Self-reported disease	15571	2709	17.4
Hearing loss	Self-reported disease	10780	1628	15.1
Urinary incontinence	Self-reported disease	16208	1361	8.4
Chronic pain	Self-reported disease	10914	1888	17.3
Angioplasty/stent	Procedure	9233	129	1.4
Hypertension	Doctor diagnosis	250	104	41.6

Appendix I

Track A and Track B: Item index and scoring of individual features

Table I.1: 40-items: and scoring of individual features

Item	Variable	Scoring
Doctor diagnosed eye disease	ph105_5	0 = None; 1 = Yes
Doctor diagnosed hypertension	ph201_01	1 = Yes; 0 = No
Doctor diagnosed high cholesterol	ph201_08	0 = No; 1 = Yes
No conditions	ph201_14	1 = None; 0 = Otherwise
Doctor diagnosed arthritis	ph301_03	1 = Yes; 0 = No
No listed conditions	ph301_17	1 = None; 0 = Otherwise
Self-reported long-term chronic/health problem	ph003	1 = Yes; 0 = No; -1 = DK/RF
Long term illness/health problem	DISlongterm	0 = No illness; 1 = Illness; 1 = Limiting illness
Neoplasms	ICD10_02	0 = No; 1 = Yes
Diseases of the blood	ICD10_03	0 = No; 1 = Yes
Endocrine/metabolic disease	ICD10_04	0 = No; 1 = Yes
Mental and behavioural disorders	ICD10_05	0 = No; 1 = Yes
Diseases of the nervous system	ICD10_06	0 = No; 1 = Yes

Continued on next page

Item	Variable	Scoring
Eye disease	ICD10_07	0 = No; 1 = Yes
Circulatory disease	ICD10_09	0 = No; 1 = Yes
Respiratory disease	ICD10_10	0 = No; 1 = Yes
Digestive system disease	ICD10_11	0 = No; 1 = Yes
Musculoskeletal disease	ICD10_13	0 = No; 1 = Yes
Genitourinary disease	ICD10_14	0 = No; 1 = Yes
Family history of diabetes	ph726_01	0 = No; 1 = Yes
Family history of high cholesterol	ph726_02	0 = No; 1 = Yes
Family history of hypertension	ph726_03	0 = No; 1 = Yes
Family history of heart disease	ph726_04	0 = No; 1 = Yes
Family history of osteoporosis	ph726_06	0 = No; 1 = Yes
Family history of alzheimer's / dementia	ph726_07	0 = No; 1 = Yes
Family history of cancer	ph726_08_to_11	1 = Yes; 0 = No
Family history of depression	ph726_12	0 = No; 1 = Yes
Family history of other condition	ph726_95	0 = No; 1 = Yes
No family history of any disease	ph726_96	0 = No; 1 = Yes
Pain medication use	ph505	1 = Yes; 0 = No; -1 = DK/RF
Cholesterol medication	ph225	1 = Yes; 0 = No; -1 = DK/RF
Antidepressant use	MDantidepressant	0 = No; 1 = Yes
Antihypertensive use	MDantihypertensives	0 = No; 1 = Yes
Self-rated chronic pain	ph501 (proxy)	1 = Yes; 0 = No; -1 = DK
Self-rated chronic pain	CHRany_pain	0 = No; 1 = Yes
Self-rated hearing loss	ph145	0 = No; 1 = Yes; -1 = DK
Self-rated urinary incontinence	ph601	0 = No; 1 = Yes; -1 = DK/RF
Angioplasty/stent	ph208	0 = No; 1 = Yes; -1 = DK
Fear of falling	ph408	0 = No; 1 = Yes; -1 = DK
Antihypertensive medication use	ph202a	1 = Yes; 0 = No; -1 = DK

Appendix J

D1 and D2: Top 20 features

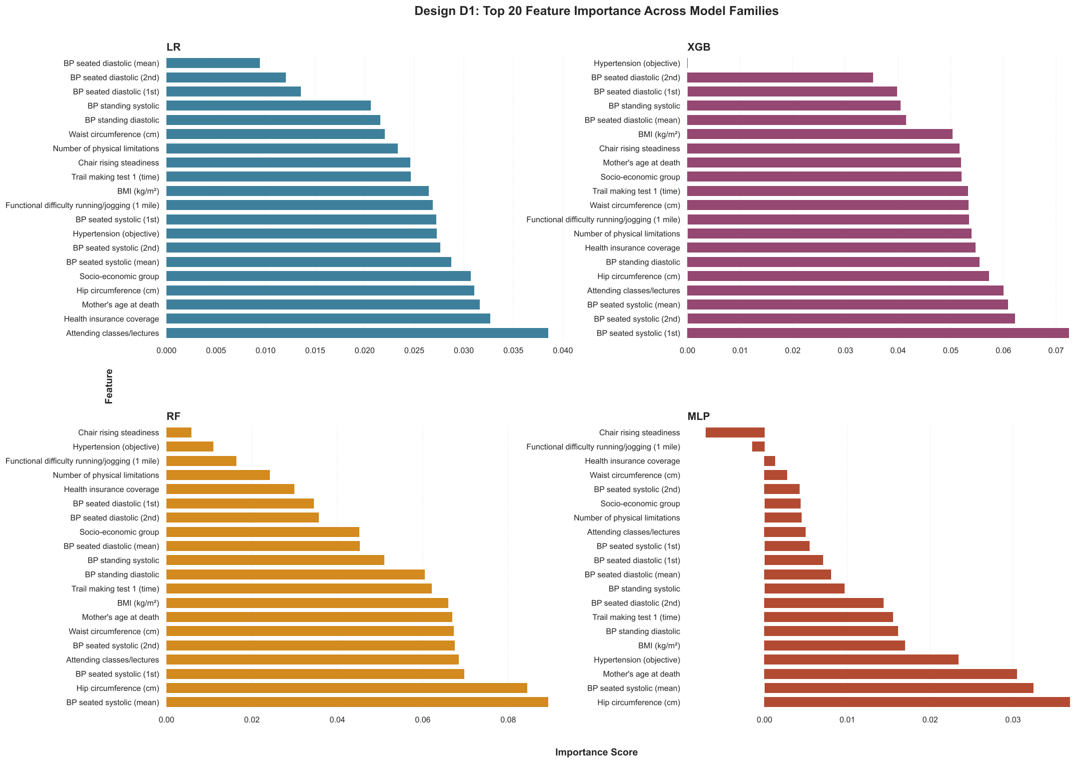

Figure J.1: Top 20 features by importance for D1.

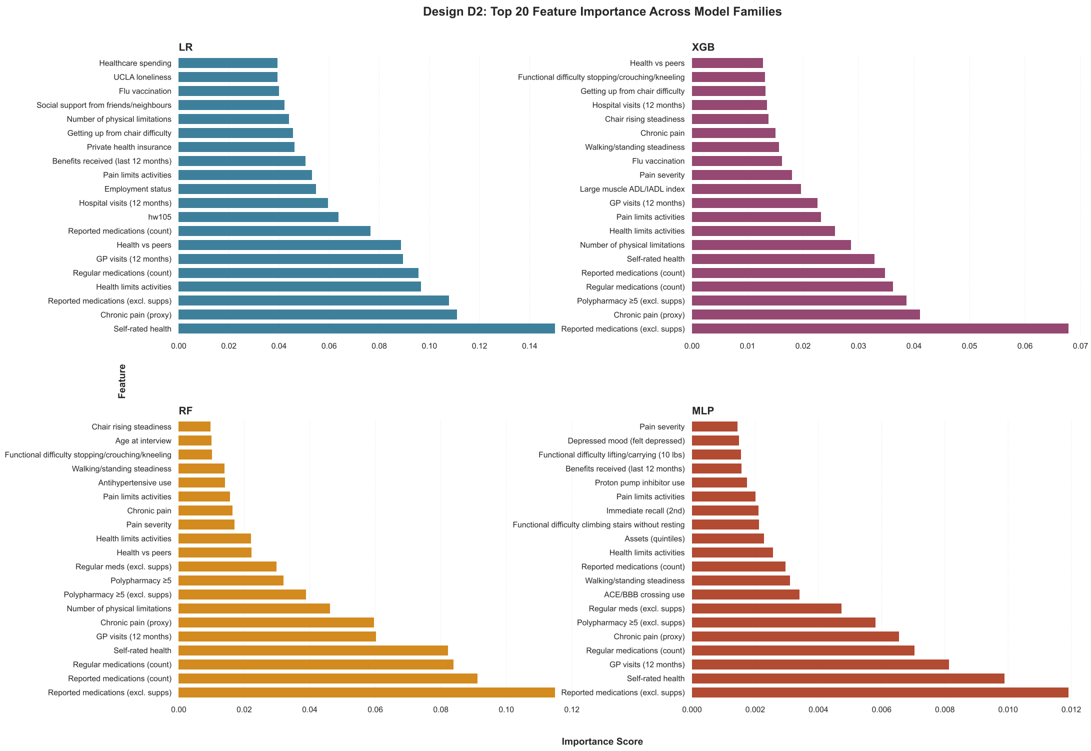

Figure J.2: Top 20 features by importance for D2.

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